Print sheet

Generated 2026-11-06 03:45. Bed 256 x 256 x 260 mm (Bambu X2D). Masses and times are rough estimates: walls and infill per material from the settings below, volume at a typical print rate plus 5 s per 0.2 mm layer, the layer time counted once per plate.

Warnings

frame_deck_ext: print bbox 246.5 x 208.0 fits the bed but not inside the 5 mm margin; printed alone, centred on its own plate

launch_guard_l_a: print bbox 250.0 x 49.0 fits the bed but not inside the 5 mm margin; printed alone, centred on its own plate

launch_guard_r_a: print bbox 250.0 x 49.0 fits the bed but not inside the 5 mm margin; printed alone, centred on its own plate

Parts

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
frame_deck_ext	PLA	1	246.5 x 208.0 x 8.0	405.4	340.5	8.53	0	no
frame_deck_3	PLA	1	228.9 x 211.3 x 8.0	358.7	302.3	7.58	0	no
frame_deck_1	PLA	1	219.9 x 211.3 x 8.0	344.8	290.3	7.28	0	no
frame_deck_4	PLA	1	228.9 x 169.3 x 8.0	295.6	249.9	6.28	0	no
reel_wall	PLA	1	201.3 x 82.0 x 213.0	294.2	291.1	8.72	1757	yes, tree supports from the plate (1757 mm2 over 45 deg)
frame_deck_2	PLA	1	219.9 x 169.3 x 8.0	284.8	241.0	6.06	0	no
frame_camera_mount	PLA	1	58.0 x 72.0 x 236.4	236.8	166.5	5.79	672	yes, tree supports from the plate (672 mm2 over 45 deg)
frame_cheek_r	PLA	1	215.0 x 220.0 x 30.0	234.7	197.7	5.13	5118	yes, tree supports from the plate (5118 mm2 over 45 deg)
launch_aim_housing	PLA	1	176.5 x 59.0 x 183.0	212.8	185.4	5.89	761	yes, tree supports from the plate (761 mm2 over 45 deg)
launch_rail_mid	PLA	1	92.5 x 68.0 x 170.0	201.1	178.6	5.63	2315	yes, tree supports from the plate (2315 mm2 over 45 deg)
drum_drum_a	PLA	1	97.1 x 97.1 x 75.0	195.6	146.5	4.17	732	yes, tree supports from the plate (732 mm2 over 45 deg)
frame_motor_leg	PLA	1	49.0 x 69.0 x 187.0	159.9	117.3	4.22	222	yes, tree supports from the plate (222 mm2 over 45 deg)
launch_rail_rear	PLA	1	180.9 x 68.0 x 76.1	150.0	135.9	3.91	4407	yes, tree supports from the plate (4407 mm2 over 45 deg)
launch_rail_front	PLA	1	131.5 x 128.0 x 48.5	146.5	137.7	3.76	7890	yes, tree supports from the plate (7890 mm2 over 45 deg)

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
frame_ebay	PLA	1	124.0 x 124.0 x 58.0	141.3	175.3	4.77	879	yes, tree supports from the plate (879 mm2 over 45 deg)
reel_cheek_i	PLA	1	63.5 x 225.5 x 22.0	115.9	102.1	2.70	13980	yes, tree supports from the plate (13980 mm2 over 45 deg)
reel_cheek_o	PLA	1	63.5 x 225.5 x 22.0	115.9	102.1	2.70	375	yes, tree supports from the plate (375 mm2 over 45 deg)
drum_drum_b	PLA	1	60.0 x 60.0 x 58.4	109.1	79.0	2.37	1818	yes, tree supports from the plate (1818 mm2 over 45 deg)
launch_stop_spring_l	PLA	1	154.9 x 51.5 x 14.0	65.8	81.6	2.13	57	no
launch_stop_spring_r	PLA	1	154.9 x 51.5 x 14.0	65.8	81.6	2.13	57	no
reel_t_bracket	PLA	1	48.0 x 195.5 x 20.0	58.6	58.3	1.59	27	no
frame_ebay_lid	PLA	1	124.0 x 124.0 x 7.0	57.6	71.4	1.83	80	no
launch_guard_l_a	PLA	1	250.0 x 49.0 x 40.0	52.4	65.0	1.90	93	no
launch_guard_r_a	PLA	1	250.0 x 49.0 x 40.0	52.4	65.0	1.90	93	no
launch_carriage	PLA	1	205.2 x 240.2 x 43.0	48.0	59.6	1.78	6453	yes, tree supports from the plate (6453 mm2 over 45 deg)
drum_bracket_a	PLA	1	104.9 x 117.0 x 6.7	37.5	37.7	0.99	32	no
frame_guard_front	PLA	1	155.0 x 104.0 x 16.4	37.1	46.0	1.26	40	no
reel_spool	PLA	1	62.0 x 62.0 x 38.0	36.1	36.1	1.16	4580	yes, tree supports from the plate (4580 mm2 over 45 deg)
frame_can_stay	PLA	1	55.0 x 55.0 x 71.0	31.0	28.4	1.20	906	yes, tree supports from the plate (906 mm2 over 45 deg)
reel_p1_leg	PLA	1	26.0 x 36.0 x 125.0	30.2	30.8	1.64	43	no
launch_stop_frame_l	PLA	1	174.6 x 53.0 x 55.1	26.3	32.7	1.20	541	yes, tree supports from the plate (541 mm2 over 45 deg)
launch_stop_frame_r	PLA	1	174.6 x 53.0 x 55.1	26.3	32.7	1.20	541	yes, tree supports from the plate (541 mm2 over 45 deg)
drum_bracket_b	PLA	1	79.0 x 117.0 x 6.0	22.0	23.4	0.62	32	no
reel_p2_leg	PLA	1	26.0 x 26.0 x 125.0	21.7	22.8	1.44	43	no
frame_guard_hood	PLA	1	122.0 x 60.0 x 14.4	21.1	26.2	0.75	84	no
drum_fork	PLA	1	87.1 x 94.1 x 83.0	21.0	25.9	1.22	1302	yes, tree supports from the plate (1302 mm2 over 45 deg)

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launch_carriage_yoke	PLA	1	43.3 x 126.6 x 28.0	18.9	23.5	0.78	1406	yes, tree supports from the plate (1406 mm2 over 45 deg)
reel_sw_bracket	PLA	1	39.5 x 63.0 x 57.9	18.6	20.6	0.92	334	yes, tree supports from the plate (334 mm2 over 45 deg)
frame_rc832_holder	PLA	1	97.0 x 21.6 x 40.0	18.4	22.8	0.84	0	no
drum_switch_block	PLA	1	26.0 x 51.0 x 24.0	18.1	16.6	0.58	216	yes, tree supports from the plate (216 mm2 over 45 deg)
launch_guard_r_b2	PLA	1	145.3 x 32.5 x 40.0	17.6	21.8	0.82	56	no
sensors_sw_rail_low_bracket	PLA	1	31.0 x 48.0 x 74.0	16.1	17.8	0.96	65	no
launch_guard_l_b2	PLA	1	145.3 x 32.5 x 40.0	14.4	17.9	0.72	33	no
drum_bracket_idler2	PLA	1	50.1 x 141.0 x 6.0	13.9	16.2	0.44	32	no
sensors_as5600_carrier	PLA	1	34.0 x 67.5 x 11.0	12.4	12.7	0.39	90	no
drum_bracket_idler	PLA	1	52.0 x 71.1 x 7.0	11.8	12.5	0.36	32	no
reel_post_t	PLA	1	26.0 x 14.0 x 119.8	11.0	12.8	1.15	43	no
reel_cup	PLA	1	48.0 x 53.4 x 18.0	10.2	12.7	0.44	1210	yes, tree supports from the plate (1210 mm2 over 45 deg)
drum_n20_cradle	PLA	1	32.0 x 53.5 x 19.0	9.6	11.9	0.43	108	yes, tree supports from the plate (108 mm2 over 45 deg)
sensors_sw_gap	PLA	1	63.5 x 43.3 x 11.5	8.0	9.9	0.33	64	no
drum_index_block	PLA	1	14.0 x 68.0 x 8.0	7.3	8.1	0.26	61	no
sensors_sw_rail_low	PLA	1	48.0 x 35.0 x 11.5	5.9	7.3	0.26	22	no
reel_sw_base	PLA	1	46.0 x 35.0 x 11.6	5.7	7.0	0.26	22	no
reel_tab_c	PLA	1	16.0 x 26.0 x 34.5	5.4	6.6	0.40	58	no
drum_dog_post	PLA	1	36.8 x 71.0 x 41.7	4.8	6.0	0.44	476	yes, tree supports from the plate (476 mm2 over 45 deg)
reel_tab_k	PLA	1	16.0 x 20.0 x 36.5	4.6	5.6	0.39	36	no
launch_aim_cap	PLA	1	45.5 x 14.0 x 19.4	4.2	5.2	0.27	0	no
launch_guard_l_c	PLA	1	32.0 x 54.5 x 26.0	3.8	4.8	0.30	3	no
launch_guard_r_c	PLA	1	32.0 x 54.5 x 26.0	3.8	4.8	0.30	3	no
launch_guard_l_b1	PLA	1	11.7 x 49.0 x 40.0	3.6	4.5	0.39	22	no
launch_guard_r_b1	PLA	1	11.7 x 49.0 x 40.0	3.6	4.5	0.39	22	no
frame_cable_anchor_1	PLA	1	16.0 x 24.0 x 10.0	3.2	4.0	0.17	141	yes, tree supports from the plate (141 mm2 over 45 deg)
frame_cable_anchor_2	PLA	1	24.0 x 16.0 x 10.0	3.2	4.0	0.17	141	yes, tree supports from the plate (141 mm2 over 45 deg)
drum_hand_lever	PLA	1	46.0 x 24.0 x 6.5	3.2	4.0	0.14	0	no
reel_spacer_1	PLA	1	10.0 x 10.0 x 33.5	2.3	2.9	0.30	0	no

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reel_spacer_2	PLA	1	10.0 x 10.0 x 33.5	2.3	2.9	0.30	0	no
reel_spacer_3	PLA	1	10.0 x 10.0 x 33.5	2.3	2.9	0.30	0	no
drum_n20_strap	PLA	1	17.9 x 53.5 x 3.0	2.2	2.7	0.09	0	no
drum_magnet_cap	PLA	1	20.0 x 20.0 x 6.5	1.5	1.9	0.09	40	no
reel_t	PLA	1	26.0 x 26.0 x 5.0	1.3	1.7	0.08	0	no
frame_ebay_esp_clamp	PLA	1	43.4 x 8.0 x 4.0	1.3	1.6	0.07	0	no
sensors_sw_gap_lever	PLA	1	40.0 x 9.0 x 8.0	1.1	1.4	0.09	32	no
sensors_sw_rail_low_lever	PLA	1	44.0 x 5.0 x 8.0	1.1	1.3	0.09	32	no
reel_sw_lever	PLA	1	42.0 x 5.0 x 8.0	1.0	1.3	0.09	32	no
reel_bar_cap_1	PLA	1	10.0 x 20.0 x 4.6	0.8	1.0	0.06	0	no
reel_bar_cap_2	PLA	1	10.0 x 20.0 x 4.6	0.8	1.0	0.06	0	no
reel_sleeve_clip	PLA	1	26.9 x 24.9 x 3.1	0.8	1.0	0.05	0	no
reel_c_clip	PLA	1	26.9 x 24.9 x 3.1	0.8	1.0	0.05	0	no
drum_sw_lever_0	PLA	1	13.7 x 47.5 x 6.0	0.8	1.0	0.07	97	no
drum_sw_lever_1	PLA	1	13.7 x 47.5 x 6.0	0.8	1.0	0.07	97	no
drum_sw_lever_2	PLA	1	13.7 x 47.5 x 6.0	0.8	1.0	0.07	97	no
sensors_as5600_strip_top	PLA	1	25.4 x 11.8 x 3.1	0.8	1.0	0.05	0	no
sensors_as5600_strip_bot	PLA	1	25.4 x 11.8 x 3.1	0.8	1.0	0.05	0	no
drum_index_lever	PLA	1	12.9 x 47.5 x 6.0	0.7	0.9	0.06	94	no
launch_sw_seated	PLA	1	41.5 x 8.4 x 6.0	0.7	0.8	0.06	24	no
launch_sw_home	PLA	1	41.5 x 8.4 x 6.0	0.7	0.8	0.06	24	no
launch_muzzle_roller	PLA	1	14.0 x 14.0 x 5.0	0.4	0.5	0.05	64	no
launch_pulley_top	PLA	1	14.0 x 14.0 x 5.0	0.4	0.5	0.05	64	no
reel_cs_clip	PLA	1	21.6 x 19.9 x 2.5	0.4	0.5	0.03	0	no
launch_gear_washer	PLA	1	18.0 x 18.0 x 2.3	0.4	0.5	0.03	0	no
launch_aim_drag_pad	PLA	1	8.8 x 24.5 x 4.0	0.3	0.4	0.04	0	no
launch_gear_clip	PLA	1	17.3 x 18.0 x 1.4	0.2	0.3	0.02	0	no
reel_cup_clip	PLA	1	14.0 x 13.7 x 2.0	0.2	0.3	0.02	0	no
launch_rail_splice_l	PLA	1	22.0 x 5.0 x 3.0	0.2	0.3	0.03	0	no
launch_rail_splice_r	PLA	1	22.0 x 5.0 x 3.0	0.2	0.3	0.03	0	no
launch_drive_washer	PLA	1	32.0 x 32.0 x 0.7	0.2	0.2	0.01	0	no
drum_shear_pin	PLA	6	12.1 x 5.0 x 4.1	0.1	0.2	0.03	0	no
reel_k_clip	PLA	1	13.5 x 12.5 x 1.5	0.1	0.1	0.01	0	no
reel_k_clip2	PLA	1	13.5 x 12.5 x 1.5	0.1	0.1	0.01	0	no
reel_kc_clip	PLA	1	13.5 x 12.5 x 1.5	0.1	0.1	0.01	0	no
launch_aim_drag_shim	PLA	1	8.6 x 7.0 x 4.0	0.1	0.1	0.03	0	no
launch_pulley_top_washer_l	PLA	1	8.8 x 8.8 x 1.5	0.1	0.1	0.01	0	no
launch_pulley_top_washer_r	PLA	1	8.8 x 8.8 x 1.5	0.1	0.1	0.01	0	no
launch_aim_idler_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
launch_aim_crankpin_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
launch_aim_lugpin_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
reel_s_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
launch_seat_clip	PLA	1	5.6 x 5.6 x 2.4	0.1	0.1	0.02	0	no
launch_muzzle_washer_l	PLA	1	6.2 x 6.2 x 0.9	0.0	0.0	0.01	0	no
launch_muzzle_washer_r	PLA	1	6.2 x 6.2 x 0.9	0.0	0.0	0.01	0	no
frame_motor_adapter	PPA-CF	1	194.0 x 194.0 x 16.0	277.9	299.3	13.41	6735	yes, tree supports from the plate (6735 mm2 over 45 deg)
launch_big_gear	PPA-CF	1	105.0 x 105.0 x 27.3	184.2	177.6	8.09	639	yes, tree supports from the plate (639 mm2 over 45 deg)
reel_p1	PPA-CF	1	161.0 x 185.0 x 14.0	89.5	111.9	5.07	0	no
drum_gear_reel	PPA-CF	1	104.0 x 103.9 x 19.0	87.9	95.9	4.39	0	no
drum_coupler	PPA-CF	1	78.0 x 77.9 x 61.5	79.6	89.3	4.40	1998	yes, tree supports from the plate (1998 mm2 over 45 deg)
reel_icr	PPA-CF	1	120.0 x 119.9 x 14.0	73.9	89.2	4.06	8349	yes, tree supports from the plate (8349 mm2 over 45 deg)
reel_p2	PPA-CF	1	154.8 x 148.3 x 6.0	68.8	86.0	3.87	0	no
reel_ic	PPA-CF	1	85.9 x 86.0 x 12.0	62.1	67.6	3.09	0	no
reel_ikr	PPA-CF	1	117.0 x 117.0 x 5.0	50.5	63.1	2.84	0	no
drum_shaft	PPA-CF	1	170.3 x 25.4 x 22.0	42.7	45.2	2.16	922	yes, tree supports from the plate (922 mm2 over 45 deg)
reel_drum	PPA-CF	1	46.0 x 46.0 x 43.0	41.0	44.5	2.27	251	yes, tree supports from the plate (251 mm2 over 45 deg)
launch_sleeve	PPA-CF	1	64.0 x 64.0 x 85.0	39.4	49.3	2.78	2907	yes, tree supports from the plate (2907 mm2 over 45 deg)
drum_collar	PPA-CF	1	60.0 x 60.0 x 26.0	38.5	44.0	2.14	2694	yes, tree supports from the plate (2694 mm2 over 45 deg)
launch_aim_crank	PPA-CF	1	62.3 x 59.6 x 38.0	28.9	35.3	1.83	985	yes, tree supports from the plate (985 mm2 over 45 deg)
launch_drive_gear	PPA-CF	1	70.0 x 69.7 x 28.8	23.9	29.9	1.53	2457	yes, tree supports from the plate (2457 mm2 over 45 deg)
drum_idler_gear	PPA-CF	1	66.0 x 65.9 x 9.3	22.0	27.5	1.29	0	no
launch_aim_output	PPA-CF	1	48.0 x 47.8 x 59.5	21.8	26.2	1.58	0	no
reel_bus	PPA-CF	1	72.0 x 71.9 x 9.5	21.0	26.3	1.23	0	no
reel_lane_c	PPA-CF	1	12.0 x 12.0 x 161.2	18.0	21.5	2.08	33	no

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launch_aim_worm	PPA-CF	1	28.0 x 28.0 x 54.5	17.1	21.3	1.33	2779	yes, tree supports from the plate (2779 mm2 over 45 deg)
reel_c	PPA-CF	1	28.0 x 28.0 x 65.4	15.8	19.8	1.33	391	yes, tree supports from the plate (391 mm2 over 45 deg)
reel_gc	PPA-CF	1	48.0 x 47.9 x 12.0	15.5	19.4	0.95	0	no
drum_dogring_launch	PPA-CF	1	62.0 x 62.0 x 10.5	15.2	19.0	0.92	123	yes, tree supports from the plate (123 mm2 over 45 deg)
launch_aim_sun	PPA-CF	1	48.0 x 47.8 x 12.0	14.6	18.3	0.89	0	no
launch_aim_lug	PPA-CF	1	30.0 x 58.0 x 27.8	13.6	16.9	0.95	995	yes, tree supports from the plate (995 mm2 over 45 deg)
reel_ti	PPA-CF	1	54.0 x 53.9 x 6.0	11.8	14.7	0.70	0	no
drum_coupler_hub	PPA-CF	1	26.0 x 26.0 x 32.5	11.7	14.6	0.88	37	no
reel_lane_k	PPA-CF	1	8.0 x 8.0 x 233.5	11.6	14.5	2.27	18	no
launch_aim_conrod	PPA-CF	1	173.0 x 17.0 x 6.0	11.3	14.2	0.67	0	no
reel_s	PPA-CF	1	62.0 x 62.0 x 48.3	11.3	14.1	0.96	11	no
reel_hub	PPA-CF	1	45.0 x 45.0 x 16.9	10.5	13.1	0.70	778	yes, tree supports from the plate (778 mm2 over 45 deg)
drum_axle	PPA-CF	1	141.0 x 10.0 x 9.0	10.5	13.1	0.64	0	no
drum_rail	PPA-CF	1	141.0 x 10.0 x 9.0	10.5	13.1	0.64	0	no
reel_icr_axle	PPA-CF	1	26.0 x 26.0 x 33.1	10.4	11.9	0.76	355	yes, tree supports from the plate (355 mm2 over 45 deg)
reel_lane_t	PPA-CF	1	8.0 x 8.0 x 202.6	10.1	12.6	1.97	4	no
reel_ik1	PPA-CF	1	54.0 x 53.9 x 5.0	9.8	12.3	0.58	0	no
reel_gk	PPA-CF	1	38.9 x 38.9 x 23.3	9.5	11.9	0.69	668	yes, tree supports from the plate (668 mm2 over 45 deg)
reel_gi	PPA-CF	1	48.0 x 47.8 x 6.0	9.1	11.3	0.55	0	no
reel_ga	PPA-CF	1	44.0 x 44.0 x 8.0	8.9	11.2	0.55	0	no
reel_cs	PPA-CF	1	16.0 x 16.0 x 42.5	8.2	9.3	0.71	10	no
reel_gt2	PPA-CF	1	44.9 x 44.9 x 6.0	7.9	9.8	0.48	0	no
reel_ss	PPA-CF	1	8.0 x 8.0 x 157.1	7.7	9.6	1.52	4	no
reel_kc	PPA-CF	1	10.0 x 10.0 x 103.2	7.7	9.6	1.14	36	no
reel_gk2	PPA-CF	1	48.0 x 47.9 x 5.0	7.6	9.5	0.46	0	no
reel_ik2	PPA-CF	1	48.0 x 47.8 x 5.0	7.6	9.4	0.45	0	no
reel_gk1	PPA-CF	1	48.0 x 47.8 x 5.0	7.4	9.3	0.45	0	no
reel_gc2	PPA-CF	1	43.8 x 43.8 x 8.0	7.4	9.2	0.47	0	no
launch_aim_bevel30	PPA-CF	1	52.6 x 52.4 x 9.7	7.0	8.8	0.46	590	yes, tree supports from the plate (590 mm2 over 45 deg)
reel_sleeve	PPA-CF	1	34.0 x 34.0 x 23.7	6.7	8.4	0.54	87	no

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reel_bcx	PPA-CF	1	44.0 x 44.0 x 7.7	6.4	8.0	0.41	337	yes, tree supports from the plate (337 mm2 over 45 deg)
reel_bar	PPA-CF	1	153.5 x 18.6 x 15.8	6.3	7.9	0.46	0	no
drum_idler	PPA-CF	1	22.0 x 21.7 x 46.0	6.0	7.6	0.66	331	yes, tree supports from the plate (331 mm2 over 45 deg)
reel_bc	PPA-CF	1	44.0 x 44.0 x 8.2	6.0	7.5	0.39	337	yes, tree supports from the plate (337 mm2 over 45 deg)
launch_aim_idler	PPA-CF	1	30.0 x 29.7 x 12.0	6.0	7.5	0.42	0	no
launch_aim_arm	PPA-CF	1	63.6 x 41.0 x 19.3	5.3	6.6	0.43	14	no
reel_rocker	PPA-CF	1	29.0 x 91.7 x 6.0	4.9	6.1	0.31	27	no
reel_hub_flange	PPA-CF	1	62.0 x 62.0 x 1.9	4.8	6.0	0.28	71	no
reel_k	PPA-CF	1	10.0 x 10.0 x 58.7	4.4	5.5	0.65	40	no
reel_swing_arm	PPA-CF	1	84.1 x 38.0 x 8.6	4.4	5.5	0.30	0	no
launch_gear_pin	PPA-CF	1	53.0 x 16.0 x 12.5	4.3	5.4	0.33	68	no
reel_gr	PPA-CF	1	33.0 x 33.0 x 6.4	3.7	4.7	0.25	0	no
reel_ga2	PPA-CF	1	33.0 x 33.0 x 5.0	3.2	4.0	0.21	0	no
reel_gkc	PPA-CF	1	33.0 x 33.0 x 5.0	3.1	3.8	0.20	0	no
reel_bk2	PPA-CF	1	30.0 x 29.7 x 7.5	2.7	3.3	0.20	166	yes, tree supports from the plate (166 mm2 over 45 deg)
reel_bkx	PPA-CF	1	30.0 x 29.7 x 7.0	2.5	3.1	0.19	166	yes, tree supports from the plate (166 mm2 over 45 deg)
reel_gt3	PPA-CF	1	27.0 x 27.0 x 6.0	2.3	2.9	0.17	0	no
reel_gs	PPA-CF	1	27.0 x 27.0 x 6.0	2.3	2.9	0.17	0	no
reel_t2	PPA-CF	1	8.0 x 8.0 x 39.0	1.9	2.4	0.38	18	no
reel_crank	PPA-CF	1	21.1 x 28.0 x 12.4	1.7	2.1	0.18	0	no
reel_ic_axle	PPA-CF	1	12.0 x 12.0 x 30.8	1.6	2.0	0.30	0	no
reel_k2	PPA-CF	1	8.0 x 8.0 x 33.2	1.6	2.0	0.32	18	no
reel_bt2	PPA-CF	1	22.0 x 22.0 x 6.8	1.2	1.5	0.11	81	no
reel_btx	PPA-CF	1	22.0 x 22.0 x 6.3	1.1	1.4	0.10	81	no
reel_bt3x	PPA-CF	1	22.0 x 22.0 x 6.3	1.1	1.4	0.10	81	no
reel_bt3	PPA-CF	1	22.0 x 22.0 x 5.8	1.0	1.3	0.10	81	no
reel_ik1_axle	PPA-CF	1	12.0 x 12.0 x 17.8	1.0	1.2	0.18	63	no
reel_ik2_axle	PPA-CF	1	12.0 x 12.0 x 17.8	1.0	1.2	0.18	63	no
reel_t3	PPA-CF	1	8.0 x 8.0 x 20.5	1.0	1.2	0.20	0	no
launch_yoke_peg_l	PPA-CF	1	20.0 x 8.0 x 7.4	1.0	1.2	0.11	0	no
launch_yoke_peg_r	PPA-CF	1	20.0 x 8.0 x 7.4	1.0	1.2	0.11	0	no
reel_ti_axle	PPA-CF	1	12.0 x 12.0 x 14.8	0.8	1.0	0.15	63	no
launch_seat_plunger	PPA-CF	1	29.5 x 15.0 x 19.6	0.8	1.0	0.18	0	no
drum_reel_washer	PPA-CF	1	35.1 x 35.1 x 1.4	0.8	0.9	0.05	0	no
reel_ikr_axle	PPA-CF	1	12.0 x 12.0 x 7.8	0.5	0.6	0.08	63	no

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
drum_n20_pinion	PPA-CF	1	9.8 x 9.5 x 8.0	0.4	0.5	0.08	0	no
launch_seat_pushrod	PPA-CF	1	16.0 x 4.8 x 3.6	0.1	0.2	0.03	0	no
reel_crank_washer	PPA-CF	1	12.0 x 12.0 x 1.2	0.1	0.2	0.02	0	no

frame_deck_ext: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 8 x M3 rover holes (it is its own drill template)
frame_deck_3: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 7 x M3 rover holes (it is its own drill template)
frame_deck_1: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template)
frame_deck_4: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 11 x M3 rover holes (it is its own drill template)
reel_wall: Rotated: the part's +y axis points up. Reel wall: foot, take-off bosses (C, K2, T2, idlers), rocker ears, stop pins, brake guides and crank boss. Print on its foot
frame_deck_2: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 8 x M3 rover holes (it is its own drill template)
frame_camera_mount: As modelled, bottom face on the bed. Kiyo X mount: its own clip grips the 16 mm edge plate (or a 1/4-20 screw), 4 x M3 x 10 into the deck
frame_cheek_r: Rotated: the part's -y axis points up. motor plate (y -10..0, front face = the spec's y 0): window for the puck adapter, 8 rim holes (M3 x 12 from the front into the puck lip), bridge to the motor leg (2 x M3 x 12), can stay and hood inserts; carved round the other subsystems
launch_aim_housing: As modelled, bottom face on the bed. Aim housing on the deck (4 x M3 x 10 from the top through its counterbored foot flanges into the deck inserts): sleeve end bushing, countershaft bushings, worm saddles, M8 wheel bolt (head on the outer boss, nut in the inner face hex pocket), engaged and swung-out swing arm stop posts
launch_rail_mid: Rotated: the part's -x axis points up. Rail mid piece: shoe track, cheeks on the sleeve, big gear pin bosses, lever seats, SW_GAP bracket pilots
drum_drum_a: As modelled, bottom face on the bed. Drum A: N20 gear 120T m0.8, brake cam, wrap segment gear + locking arc, hex spigot
frame_motor_leg: As modelled, bottom face on the bed. motor support leg (prints upright): 4 x M3 x 10 into the deck, 2 x M3 x 12 along y into the bridge behind the motor plate
launch_rail_rear: As modelled, bottom face on the bed. Rail rear piece: shoe track, walls, tether fork and two pulleys behind the rear end
launch_rail_front: As modelled, bottom face on the bed. Rail front piece: shoe track end, spring anchor posts with straps, muzzle tie
frame_ebay: As modelled, bottom face on the bed. electronics bay: ESP32-S3 cradle (USB slot), BTS7960 on standoffs with vents, DRV8833, MP1584, HX711 pockets, fuse cradle, wire exits with tie anchors
reel_cheek_i: Rotated: the part's +y axis points up. Inboard cheek: spool shaft 608, P2 bolts to its front edge. Print flat
reel_cheek_o: Rotated: the part's +y axis points up. Outboard cheek: spool shaft 608, P2 bolts to its front edge. Print flat
drum_drum_b: As modelled, bottom face on the bed. Drum B: shift groove barrel, Gray code bands A B C, hand lever hex boss
launch_stop_spring_l: Rotated: the part's -z axis points up. Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_stop_spring_r: Rotated: the part's -z axis points up. Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
reel_t_bracket: Rotated: the part's +y axis points up. T bracket: bushings for T3 and the spool shaft behind the 45T pair. Print flat
frame_ebay_lid: Rotated: the part's -z axis points up. bay lid: grille over the BTS7960, 4 x M3 into the corner bosses
launch_guard_l_a: Rotated: the part's -y axis points up. Spring guard left rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_a: Rotated: the part's +y axis points up. Spring guard right rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_carriage: Rotated: the part's +y axis points up. Carriage: rack, shoes, dart cradle, pusher with tether hole, spring arms and pegs, SW feet
drum_bracket_a: As modelled, bottom face on the bed. Drum bracket A: axle, shift rail, N20 ear, 2 x M3 to the deck
frame_guard_front: Rotated: the part's -x axis points up. front gear guard between the drum gears and the bay, M3 x 10 into the deck
reel_spool: Rotated: the part's -z axis points up. Free storage spool: 20 m of 1 mm Dyneema on a dia 40 core, d62 flanges, keyed on the spool shaft
frame_can_stay: Rotated: the part's -y axis points up. can stay: post on the motor plate (2 x M3 x 12 into plate inserts), cable tie or hose clamp round the can through the cradle slots; reprint with the adapter
reel_p1_leg: As modelled, bottom face on the bed. P1 leg: from the deck up to P1; lane T runs through its bushing. Print on its side
launch_stop_frame_l: Rotated: the part's -y axis points up. Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_frame_r: Rotated: the part's +y axis points up. Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
drum_bracket_b: As modelled, bottom face on the bed. Drum bracket B: axle, shift rail, 2 x M3 to the deck

reel_p2_leg: As modelled, bottom face on the bed. P2 leg: from the deck up to P2; lane T runs through its bushing. Print on its side
frame_guard_hood: Rotated: the part's -z axis points up. gear hood over the reel and bus gears, M3 x 10 through its lugs into the motor plate
drum_fork: As modelled, bottom face on the bed. Compliant shift fork: thin yoke is the spring (about 1.5 N/mm), follower pin in groove 1, contact tab
launch_carriage_yoke: As modelled, bottom face on the bed. Bolted yoke: both spring arms (hit the stop, d8.1 holes for the 8 mm steel rod spring pegs, D5 holes at y +-56 empty), cross bar, keys, SW_SEATED block, rigid SW_HOME foot. Prints on the peg boss bottoms and key plates (w 62)
reel_sw_bracket: As modelled, bottom face on the bed. SW_ARM bracket: screwed to the wall below the gears. Print flat
frame_rc832_holder: As modelled, bottom face on the bed. RC832 holder: receiver stands antenna-end up in the slot, leads out low, 2 x M3 x 10
drum_switch_block: As modelled, bottom face on the bed. Code switch block: 3 lever feet, 3 M3 nut contacts (SW_DRUM_A, B, C)
launch_guard_r_b2: Rotated: the part's +y axis points up. Spring guard right mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 2 wall brackets (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
sensors_sw_rail_low_bracket: As modelled, bottom face on the bed. SW_RAIL_LOW bracket on the aim housing foot extension: an M3x10 into the extension and its own deck screw, an M3x16 from the top through the counterbored foot and the extension into a deck insert
launch_guard_l_b2: Rotated: the part's -y axis points up. Spring guard left mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
drum_bracket_idler2: As modelled, bottom face on the bed. Wrap idler second journal post (motor end), 2 x M3 to the deck
sensors_as5600_carrier: Rotated: the part's -y axis points up. AS5600 carrier: plate with 4 pegs through the board holes, column screwed to the aim housing spine
drum_bracket_idler: As modelled, bottom face on the bed. Wrap idler journal post, 2 x M3 to the deck
reel_post_t: As modelled, bottom face on the bed. Lane T front post on the deck extension. Print on its foot
reel_cup: As modelled, bottom face on the bed. Wrap cup (Harry v7, ring hole turned axial): ring at the bottom at home, the line passes straight through it. Print open end down
drum_n20_cradle: As modelled, bottom face on the bed. N20 cradle on the bracket A ear: pivot + swing screw (hand fallback), SW_N20 contact
sensors_sw_gap: As modelled, bottom face on the bed. SW_GAP printed contact base (cad.lib.switch, coupon 2 geometry): with its arm screwed to the right rail cheek outer face below the big gear, closed by the big gear cam bump (gear in its gap)
drum_index_block: As modelled, bottom face on the bed. SW_INDEX block: 2 x M3 x 12 through the idler post 2 strut, M3 nut contact
sensors_sw_rail_low: As modelled, bottom face on the bed. SW_RAIL_LOW printed contact base (cad.lib.switch, coupon 2 geometry): on its bracket on the aim housing foot, closed by the crank cam at the LOW dead point
reel_sw_base: As modelled, bottom face on the bed. SW_ARM contact base (printed switch, M3 head on M3 nut)
reel_tab_c: Rotated: the part's -y axis points up. Lane C bushing tab: screwed to the wall front. Print on its flange
drum_dog_post: Rotated: the part's -z axis points up. Dog contact post on the shift rail: SW_DOG_R and SW_DOG_L screws (collar fully engaged)
reel_tab_k: Rotated: the part's -y axis points up. Lane K bushing tab: screwed to the wall front. Print on its flange
launch_aim_cap: Rotated: the part's -z axis points up. Worm shaft saddle cap, 4 x M3x30 through the legs into 13 deep pilots in the saddles
launch_guard_l_c: Rotated: the part's -y axis points up. Spring guard left front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_guard_r_c: Rotated: the part's +y axis points up. Spring guard right front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_guard_l_b1: Rotated: the part's -y axis points up. Spring guard left mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b1: Rotated: the part's +y axis points up. Spring guard right mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
frame_cable_anchor_1: As modelled, bottom face on the bed. cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
frame_cable_anchor_2: As modelled, bottom face on the bed. cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
drum_hand_lever: As modelled, bottom face on the bed. Hand lever on the drum hex boss (fallback: swing the N20 out first)
reel_spacer_1: Rotated: the part's -x axis points up. Spacer tube between P1 and P2 (M3 through)
reel_spacer_2: Rotated: the part's -x axis points up. Spacer tube between P1 and P2 (M3 through)
reel_spacer_3: Rotated: the part's -x axis points up. Spacer tube between P1 and P2 (M3 through)
drum_n20_strap: As modelled, bottom face on the bed. End plate closing the N20 cradle, 2 x M3 x 8
drum_magnet_cap: Rotated: the part's -z axis points up. AS5600 magnet cap on the spigot end (6 x 2.5 diametric or edge-on disc), M3 x 6 onto the flat
reel_t: As modelled, bottom face on the bed. Thrust ring: the rocker pushes it onto the rear 608 inner race
frame_ebay_esp_clamp: As modelled, bottom face on the bed. ESP32 hold-down over the module end, 2 x M3 x 8 into the cradle posts
sensors_sw_gap_lever: Rotated: the part's +y axis points up. SW_GAP lever, t 1.4 mm, print on its side so layers run along the beam
sensors_sw_rail_low_lever: Rotated: the part's +y axis points up. SW_RAIL_LOW lever, t 1.4 mm, print on its side so layers run along the beam
reel_sw_lever: Rotated: the part's +y axis points up. SW_ARM lever: rides the cup gear's raised track (contact closed) and drops into its home notch (open). Print on its side
reel_bar_cap_1: Rotated: the part's -y axis points up. Bar guide cap: closes the U guide over the bar. Print flat

reel_bar_cap_2: Rotated: the part's -y axis points up. Bar guide cap: closes the U guide over the bar. Print flat
reel_sleeve_clip: As modelled, bottom face on the bed. C-clip on the sleeve behind the bus gear hub
reel_c_clip: As modelled, bottom face on the bed. C-clip behind G_c
drum_sw_lever_0: As modelled, bottom face on the bed. Code switch lever A: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_1: As modelled, bottom face on the bed. Code switch lever B: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_2: As modelled, bottom face on the bed. Code switch lever C: flexure, round nub, M3 x 6 contact screw
sensors_as5600_strip_top: Rotated: the part's +y axis points up. AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
sensors_as5600_strip_bot: Rotated: the part's +y axis points up. AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
drum_index_lever: As modelled, bottom face on the bed. SW_INDEX lever on the 6-bump coupler cam (D12 fallback encoder)
launch_sw_seated: Rotated: the part's +y axis points up. SW_SEATED printed flex lever on the left rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
launch_sw_home: Rotated: the part's +y axis points up. SW_HOME printed flex lever on the right rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
launch_muzzle_roller: As modelled, bottom face on the bed. Lip roller at the carriage front: the tether bends down over it in tow and retrieval (groove r 6 for the 1 mm line), 693ZZ inside
launch_pulley_top: As modelled, bottom face on the bed. Rail-rear tether pulley (D50, the line runs under it), groove r 6, 693ZZ inside
reel_cs_clip: As modelled, bottom face on the bed. C-clip behind the C gear
launch_gear_washer: As modelled, bottom face on the bed. Spacer washer: big gear hub to the left cheek
launch_aim_drag_pad: As modelled, bottom face on the bed. Flexure drag pad: cantilever finger t 2, tip bump 0.45 into the sleeve d20 (strain 0.43 %, about 2.3 N), root block pushed axially into the arm hub pocket
launch_gear_clip: As modelled, bottom face on the bed. C clip on the big gear pin groove
reel_cup_clip: As modelled, bottom face on the bed. Harry's v7 C-clip on the cup shaft neck in front of the cup
launch_rail_splice_l: As modelled, bottom face on the bed. Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_rail_splice_r: As modelled, bottom face on the bed. Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_drive_washer: As modelled, bottom face on the bed. Spacer washer: drive gear hub to the left cheek
drum_shear_pin: Rotated: the part's -z axis points up. Torque limiter shear pin d4 (about 5 N m), prints lying; print 6, 5 are spares
reel_k_clip: As modelled, bottom face on the bed. C-clip holding the cup gear on the cup shaft
reel_k_clip2: As modelled, bottom face on the bed. C-clip in front of the riser gear
reel_kc_clip: As modelled, bottom face on the bed. C-clip behind P2
launch_aim_drag_shim: As modelled, bottom face on the bed. Preload shim behind the drag pad root block (nominal; print up to 0.25 thicker for more drag, strain stays under 0.7 %)
launch_pulley_top_washer_l: Rotated: the part's -z axis points up. Printed top-hat thrust washer between fork plate and 693ZZ inner race
launch_pulley_top_washer_r: Rotated: the part's -z axis points up. Printed top-hat thrust washer between fork plate and 693ZZ inner race
launch_aim_idler_clip: As modelled, bottom face on the bed. Printed C clip on the arm pin, holds the idler
launch_aim_crankpin_clip: As modelled, bottom face on the bed. Printed C clip on the crank pin, inboard of the conrod
launch_aim_lugpin_clip: As modelled, bottom face on the bed. Printed C clip on the lug pin, outboard of the conrod
reel_s_clip: As modelled, bottom face on the bed. C-clip holding the reel gear
launch_seat_clip: Rotated: the part's -z axis points up. Head cap on the plunger pin (rests on the yoke block)
launch_muzzle_washer_l: Rotated: the part's -z axis points up. Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_muzzle_washer_r: Rotated: the part's -z axis points up. Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
frame_motor_adapter: Rotated: the part's -z axis points up. puck adapter: slotted M6 holes (+-4 mm) for the motor bosses, rear lip with 8 M3 inserts for the rim screws; reprint when the motor is measured
launch_big_gear: Rotated: the part's -z axis points up. Big gear m2.5: sector layer 27 of 40 teeth (rack, v10 release teeth) + full 40T layer (drive), SW_GAP cam bump, on the rail pin
reel_p1: Rotated: the part's -x axis points up. Reel unit front plate P1 with the sprag #2 seat and the capstan bushing. Print front face down
drum_gear_reel: As modelled, bottom face on the bed. REEL output gear m2 50T face 12 (D20), free on its sleeve, dog face toward the collar
drum_coupler: As modelled, bottom face on the bed. Coupler: 5 N m shear pin limiter (D8), bus gear 50T m1.5 (D25), SW_INDEX cam, REEL journal, hex socket; prints hub end down, tree support under the bus gear ring
reel_icr: Rotated: the part's -z axis points up. I_c m2 z58: idler from lane C to the capstan, sprag #2 pressed in its hub
reel_p2: Rotated: the part's -x axis points up. Reel unit rear plate P2: cup shaft and lane K journals, I_k axle. Print flat
reel_ic: As modelled, bottom face on the bed. REEL idler m2 z41: from the REEL gear to the C gear
reel_ikr: As modelled, bottom face on the bed. I_k m1.5 z76: idler from lane K to the cup shaft
drum_shaft: Rotated: the part's +y axis points up. Main shaft: hex 22 AF, d12 spigot into the launch sleeve; prints on a hex flat (support under the spigot)
reel_drum: As modelled, bottom face on the bed. Capstan drum dia 40 x 40 (Harry v7), keyed on the tube. Print flange down
launch_sleeve: Rotated: the part's -z axis points up. Launch sleeve d20: dog ring flange, integral key ribs (sprag #1, splitter sun), journals
drum_collar: As modelled, bottom face on the bed. Dog collar: hex bore, 4 back-cut roofed dogs each face, fork groove (bridges over the lower dogs)

launch_aim_crank: As modelled, bottom face on the bed. Worm wheel m2 z20 face 16 = crank gear, crank arm r30 and d8 crank pin in one part, two 608 inside
launch_drive_gear: As modelled, bottom face on the bed. Drive gear m2.5 26T, sprag #1 inside on an integral key rib
drum_idler_gear: As modelled, bottom face on the bed. Wrap idler output gear 42T m1.5 (D bore) to the cad-reel cup ring 24T: 7 cup turns
launch_aim_output: As modelled, bottom face on the bed. Splitter output m1.5 z30 face 12 with its d8 countershaft (D flat)
reel_bus: Rotated: the part's -z axis points up. Bus gear m1.5 z46 (free on the sleeve): driven by the drum's bus gear, drives the swing idler. Print face down
reel_lane_c: As modelled, bottom face on the bed. Lane C (PPA-CF d12): the capstan drive along the deck to the reel unit. Print standing
launch_aim_worm: As modelled, bottom face on the bed. Worm m2 pd24 single start (lead angle 4.8 deg, self-locking) + d8 shaft + bevel z15
reel_c: As modelled, bottom face on the bed. Capstan tube C: key rib in the sprag and the drum, D flat for G_c. Print standing
reel_gc: As modelled, bottom face on the bed. C gear m2 z22: driven by the REEL gear (50T) through the 41T idler
drum_dogring_launch: Rotated: the part's -z axis points up. LAUNCH dog ring: 3 slots for the launch sleeve lugs, 3 x M3 from the sleeve side
launch_aim_sun: As modelled, bottom face on the bed. Splitter sun m1.5 z30 face 12, keyed on the launch sleeve
launch_aim_lug: Rotated: the part's +y axis points up. Lug bracket (PPA-CF), 2 x M3 into the rail wall, d8 lug pin at (u 115.2, w 0)
reel_ti: As modelled, bottom face on the bed. Take-up idler m1.5 z34
drum_coupler_hub: As modelled, bottom face on the bed. Coupler hub on the wiper shaft (bore parametric: WIPER), M8 nut inside, shear pin hole
reel_lane_k: As modelled, bottom face on the bed. Lane K (PPA-CF d8): the cup drive along the deck to the reel unit. Print standing
launch_aim_conrod: Rotated: the part's +y axis points up. Conrod 156 between pin centres (PPA-CF), printed C clips on both pins
reel_s: Rotated: the part's -z axis points up. Reel shaft: 608 journals, head = clutch disk, D flat for the reel gear. Print head down
reel_hub: As modelled, bottom face on the bed. Clutch hub: two 608s on the reel shaft, 40T teeth drive T2 (and the spool 1:1). Print rear face down
drum_axle: Rotated: the part's -x axis points up. Fixed drum axle d10, pressed into brackets A and B
drum_rail: Rotated: the part's -x axis points up. Shift rail d10, pressed into brackets A and B
reel_icr_axle: As modelled, bottom face on the bed. I_c axle (PPA-CF d20, keyed): fixed in P1 (D flat), carries sprag #2's inner race
reel_lane_t: As modelled, bottom face on the bed. Lane T (PPA-CF d8): the take-up drive along the deck to the spool. Print standing
reel_ik1: As modelled, bottom face on the bed. Cup riser idler m1.5 z34 on its axle pin in the wall
reel_gk: As modelled, bottom face on the bed. Cup gear m1.5 z24: driven by the drum's wrap idler; the raised track on its back holds SW_ARM closed, the notch at home lets it open
reel_gi: As modelled, bottom face on the bed. Swing idler m1.5 z30: meshes the reel gear only on forward wiper turns
reel_ga: As modelled, bottom face on the bed. G_a m2 z20 on lane C
reel_cs: Rotated: the part's -z axis points up. Countershaft C (PPA-CF d16): bevel in front of the wall, wall bushing, 22T gear behind
reel_gt2: As modelled, bottom face on the bed. T2 gear m1.5 z28: meshes the clutch hub 1:1
reel_ss: As modelled, bottom face on the bed. Spool shaft (PPA-CF d8): 45T gear, bushing in the T bracket, 608s in both cheeks, the spool on its flat
reel_kc: As modelled, bottom face on the bed. Rear cup shaft K': cup in front, G_k behind, journal in P2. Print standing
reel_gk2: As modelled, bottom face on the bed. K2 riser gear m1.5 z30
reel_ik2: As modelled, bottom face on the bed. Cup riser idler m1.5 z30 on its axle pin in the wall
reel_gk1: As modelled, bottom face on the bed. Cup riser gear m1.5 z30 on the cup shaft in front of the wall (1:1 riser to K2)
reel_gc2: As modelled, bottom face on the bed. G_c m2 z20 on the capstan tube
launch_aim_bevel30: As modelled, bottom face on the bed. Bevel m1.5 z30 on the countershaft D flat, drives the z15 2:1
reel_sleeve: Rotated: the part's -z axis points up. Stationary sleeve at the old capstan axis: c1 turns on it, the cup shaft inside. Print flange down
reel_bcx: As modelled, bottom face on the bed. Lane C bevel m2 z20 (meshes the C bevel)
reel_bar: As modelled, bottom face on the bed. Brake bar: cad-drum's track 2 cam pushes its nose, its left end turns the crank. Print flat
drum_idler: As modelled, bottom face on the bed. Wrap idler shaft: pinion 10T m1.25 (locked at home by the drum lock arc), lock cam, two journals
reel_bc: As modelled, bottom face on the bed. C bevel m2 z20 (1:1 into lane C)
launch_aim_idler: As modelled, bottom face on the bed. Splitter idler m1.5 z18 face 12 on the arm pin
launch_aim_arm: Rotated: the part's -z axis points up. Swing arm on the sleeve, dragged by the printed PLA flexure pad in its hub; printed idler pin with a printed C clip groove
reel_rocker: As modelled, bottom face on the bed. Brake rocker: flexible PPA-CF lever = clutch spring, pushed by the crank. Print flat
reel_hub_flange: As modelled, bottom face on the bed. Friction disc: keeps the front 608 in, carries the clutch friction ring. Print flat
reel_k: Rotated: the part's -z axis points up. Cup shaft K: through the sleeve, cup gear behind the wall, riser gear in front. Print standing
reel_swing_arm: As modelled, bottom face on the bed. Swing arm: rides on the bus gear hub, carries the idler into mesh on forward wiper turns. Print flat
launch_gear_pin: Rotated: the part's -x axis points up. Snap pin d10 for the big gear, C clip; prints lying on its flat
reel_gr: As modelled, bottom face on the bed. Reel gear m1.5 z20: turns the clutch disk 2.5 x wiper speed
reel_ga2: As modelled, bottom face on the bed. G_a' m1.5 z20 on lane K
reel_gkc: As modelled, bottom face on the bed. G_k m1.5 z20 on the cup shaft K'
reel_bk2: As modelled, bottom face on the bed. K bevel m1.5 z18 on K2 (1:1 into lane K)
reel_bkx: As modelled, bottom face on the bed. Lane K bevel m1.5 z18 (meshes the K2 bevel)

reel_gt3: As modelled, bottom face on the bed. T3 gear m1.5 z16
reel_gs: As modelled, bottom face on the bed. Spool shaft gear m1.5 z16
reel_t2: Rotated: the part's -z axis points up. T2 stub (PPA-CF d8): wall bushing, 40T gear and the T bevel
reel_crank: Rotated: the part's -y axis points up. Brake bell crank: the bar turns it, arm B pushes the rocker leg. Print on its flat side
reel_ic_axle: As modelled, bottom face on the bed. REEL idler axle pin (PPA-CF d8): pressed into the wall from behind, head behind the idler
reel_k2: Rotated: the part's -z axis points up. K2 stub (PPA-CF d8): wall bushing, riser gear and the K bevel
reel_bt2: As modelled, bottom face on the bed. T bevel m1 z20 on T2 (1:1 into lane T)
reel_btx: As modelled, bottom face on the bed. Lane T front bevel m1 z20 (meshes the T2 bevel)
reel_bt3x: As modelled, bottom face on the bed. Lane T rear bevel m1 z20 (meshes T3 in front of its axis)
reel_bt3: As modelled, bottom face on the bed. T3 bevel m1 z20
reel_ik1_axle: As modelled, bottom face on the bed. Idler axle pin (PPA-CF d8): pressed into the wall from the front, head in front of the idler
reel_ik2_axle: As modelled, bottom face on the bed. Idler axle pin (PPA-CF d8): pressed into the wall from the front, head in front of the idler
reel_t3: As modelled, bottom face on the bed. T3 stub (PPA-CF d8): rear T bevel and the 45T gear
launch_yoke_peg_l: Rotated: the part's +x axis points up. Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
launch_yoke_peg_r: Rotated: the part's +x axis points up. Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
reel_ti_axle: As modelled, bottom face on the bed. Take-up idler axle pin (PPA-CF d8): pressed into the T bracket, head on the inboard side of the idler
launch_seat_plunger: As modelled, bottom face on the bed. SW_SEATED plunger: foot pad + stem with a 45 deg chamfer, clip groove on top
drum_reel_washer: As modelled, bottom face on the bed. REEL gear retaining washer on the coupler journal end, 2 x M3 x 6
reel_ikr_axle: As modelled, bottom face on the bed. I_k axle pin (PPA-CF d8): pressed into P2, head in front of the idler
drum_n20_pinion: As modelled, bottom face on the bed. N20 pinion 10T m0.8, D bore pressed on the N20 shaft
launch_seat_pushrod: Rotated: the part's -z axis points up. SW_SEATED pushrod d3.2 with the stem slot
reel_crank_washer: As modelled, bottom face on the bed. Printed washer: keeps the crank on its boss

Plates

plate	material	objects	mass g	time h
plate_1_PLA.3mf	PLA	3	94	2.6
plate_2_PLA.3mf	PLA	4	312	8.1
plate_3_PLA.3mf	PLA	15	300	8.3
plate_4_PLA.3mf	PLA	1	290	7.3
plate_5_PLA.3mf	PLA	3	353	8.9
plate_6_PLA.3mf	PLA	2	343	8.7
plate_7_PLA.3mf	PLA	13	626	17.1
plate_8_PLA.3mf	PLA	19	540	15.1
plate_9_PLA.3mf	PLA	8	379	10.7
plate_10_PLA.3mf	PLA	10	239	6.5
plate_11_PLA.3mf	PLA	23	561	15.3
plate_12_PLA.3mf	PLA	11	101	3.0
plate_13_PLA.3mf	PLA	1	340	8.5
plate_14_PLA.3mf	PLA	1	65	1.9
plate_15_PLA.3mf	PLA	1	65	1.9
plate_16_PPA-CF.3mf	PPA-CF	8	436	19.7
plate_17_PPA-CF.3mf	PPA-CF	7	285	13.1
plate_18_PPA-CF.3mf	PPA-CF	16	337	16.6
plate_19_PPA-CF.3mf	PPA-CF	11	466	21.9
plate_20_PPA-CF.3mf	PPA-CF	17	249	11.5
plate_21_PPA-CF.3mf	PPA-CF	28	168	8.2

Totals on the plates

material	objects	plates	mass g	filament m	time h
PLA	115	15	4609	1545.2	123.9
PPA-CF	87	6	1943	646.1	90.9

Recommended settings

PLA

- 0.4 mm nozzle, 0.2 mm layers
- 4 walls, 5 top and 5 bottom layers
- 40 % gyroid infill
- Nozzle 220 C, bed 55 to 60 C, textured PEI plate
- Door open or top vented

PPA-CF

- Hardened steel nozzle (0.4 or 0.6 mm), never a brass nozzle
- 0.2 mm layers
- 6 walls, 6 top and 6 bottom layers
- 60 % gyroid infill
- Dry the filament first (80 C for 8 h or more) and print from a dry box
- Enclosure closed, chamber heated if the printer can
- Nozzle 290 to 320 C, bed 100 to 110 C (check the spool label)

plate_1_PLA.3mf

PLA, 3 objects, about 94 g and 2.6 h.

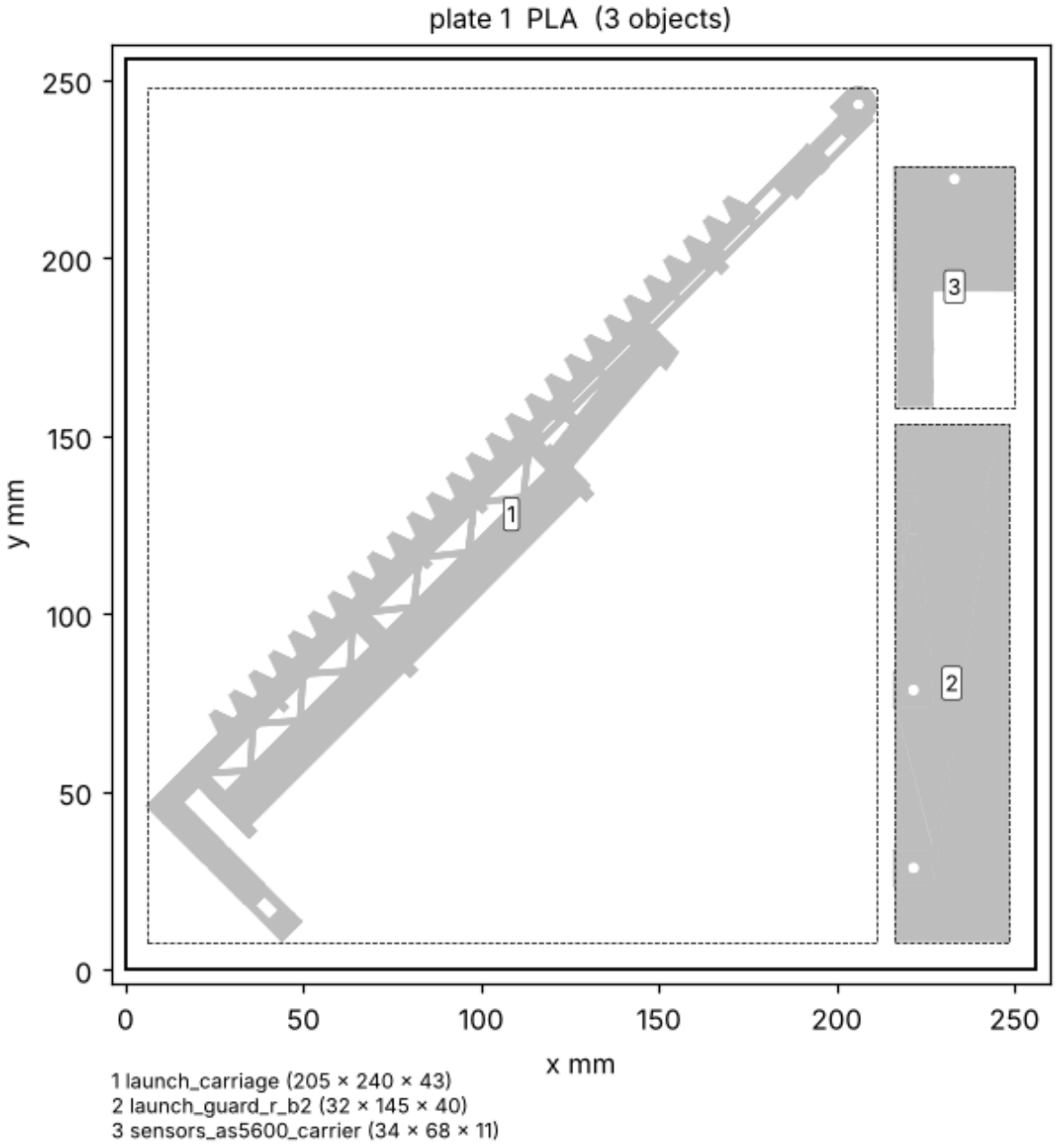
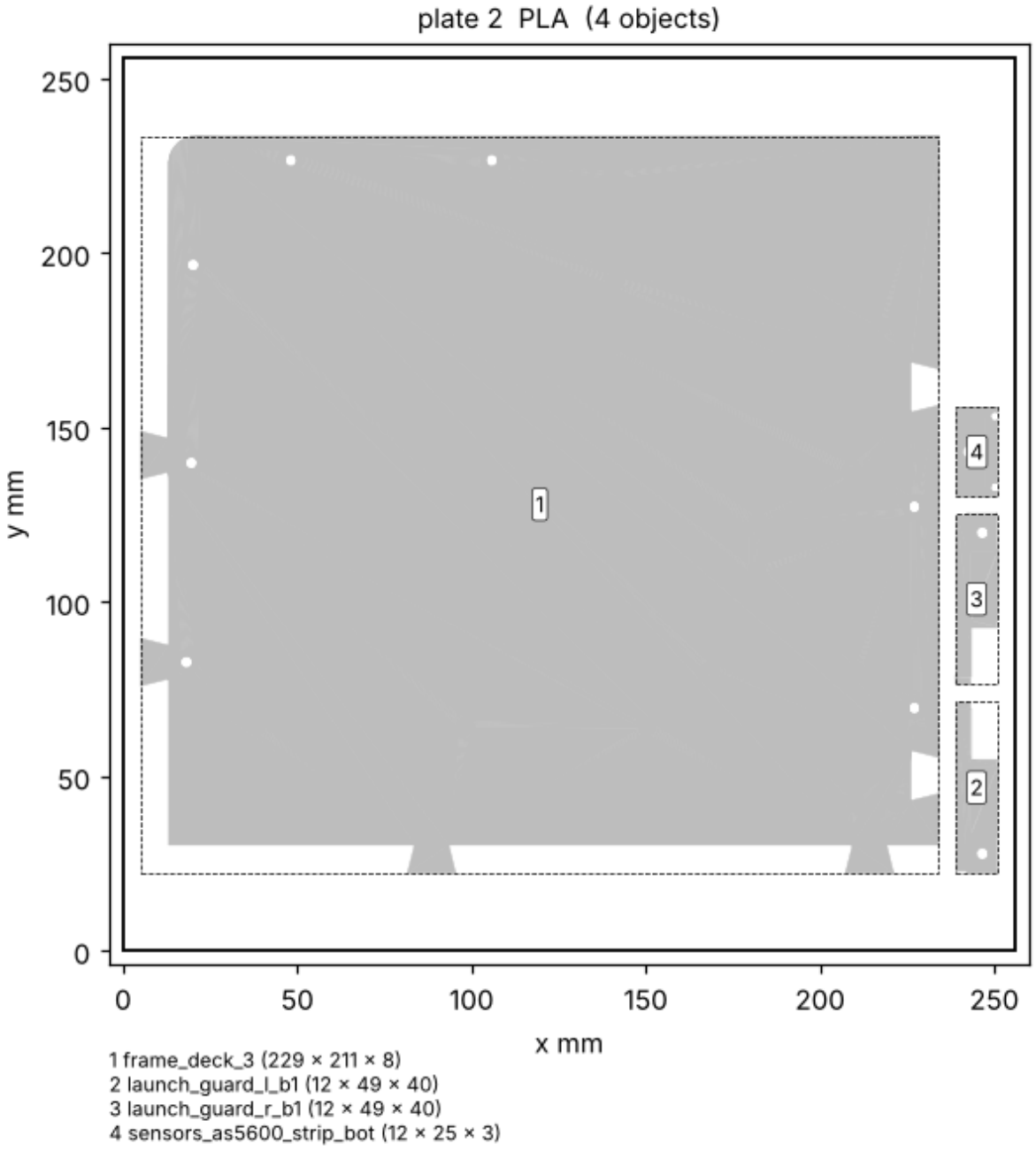


plate_2_PLA.3mf

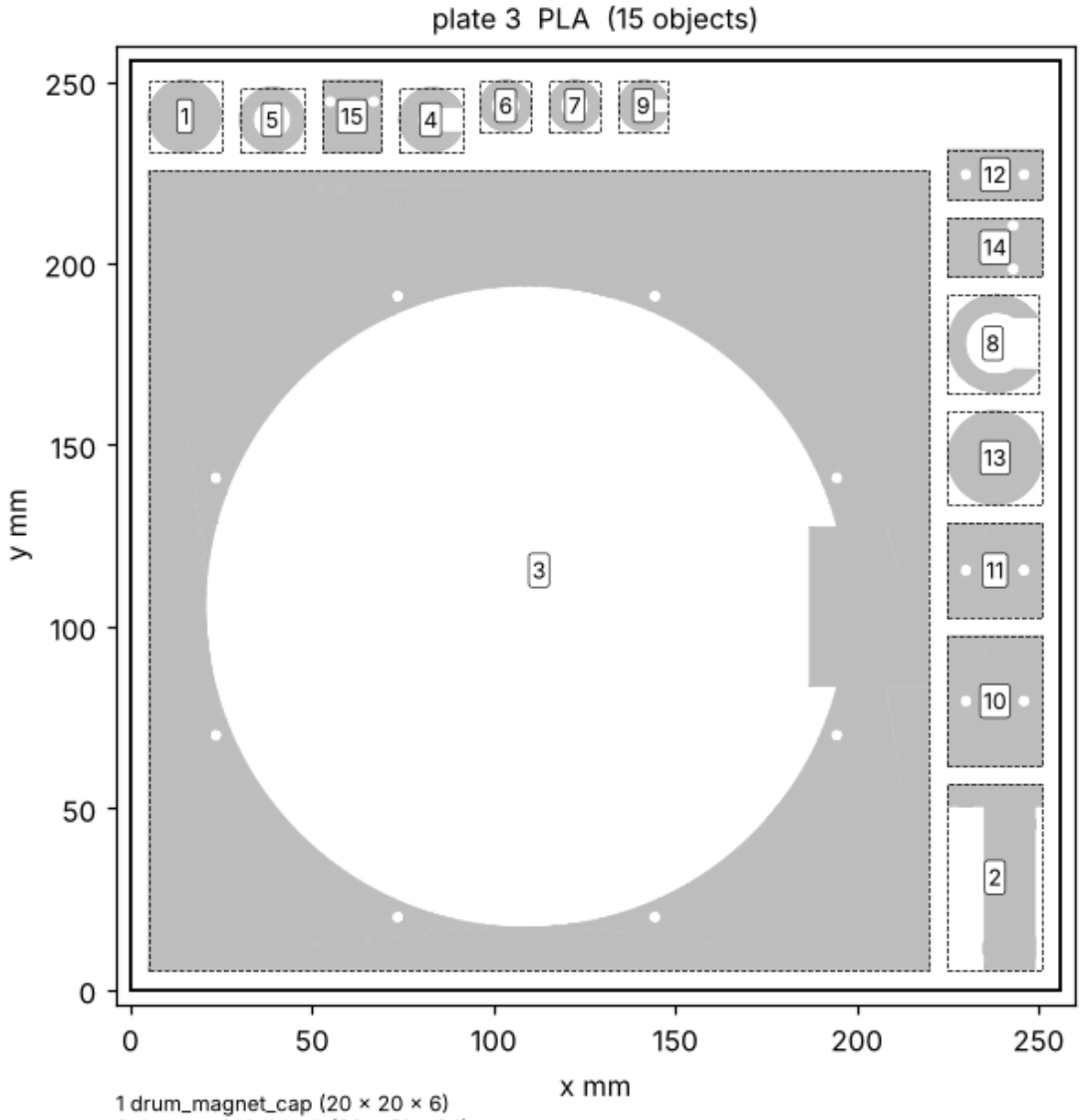
PLA, 4 objects, about 312 g and 8.1 h.



1 frame_deck_3 (229 × 211 × 8)
2 launch_guard_l_b1 (12 × 49 × 40)
3 launch_guard_r_b1 (12 × 49 × 40)
4 sensors_as5600_strip_bot (12 × 25 × 3)

plate_3_PLA.3mf

PLA, 15 objects, about 300 g and 8.3 h.



1 drum_magnet_cap (20 × 20 × 6)
2 drum_switch_block (26 × 51 × 24)
3 frame_cheek_r (215 × 220 × 30)
4 launch_gear_clip (17 × 18 × 1)
5 launch_gear_washer (18 × 18 × 2)
6 launch_muzzle_roller (14 × 14 × 5)
7 launch_pulley_top (14 × 14 × 5)
8 reel_c_clip (25 × 27 × 3)
9 reel_cup_clip (14 × 14 × 2)
10 reel_p1_leg (26 × 36 × 125)
11 reel_p2_leg (26 × 26 × 125)
12 reel_post_t (26 × 14 × 120)
13 reel_t (26 × 26 × 5)
14 reel_tab_c (26 × 16 × 34)
15 reel_tab_k (16 × 20 × 36)

plate_4_PLA.3mf

PLA, 1 objects, about 290 g and 7.3 h.

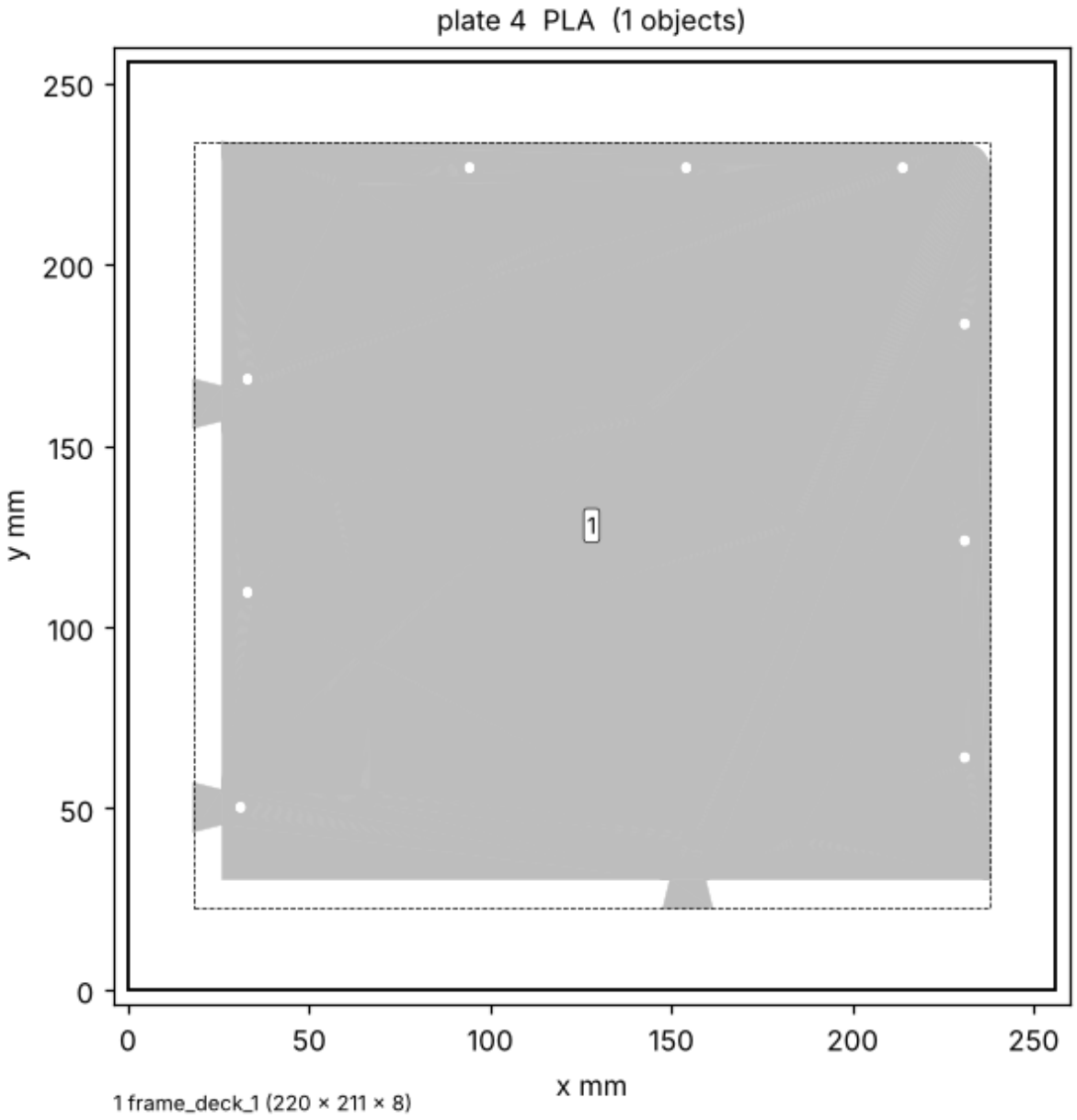
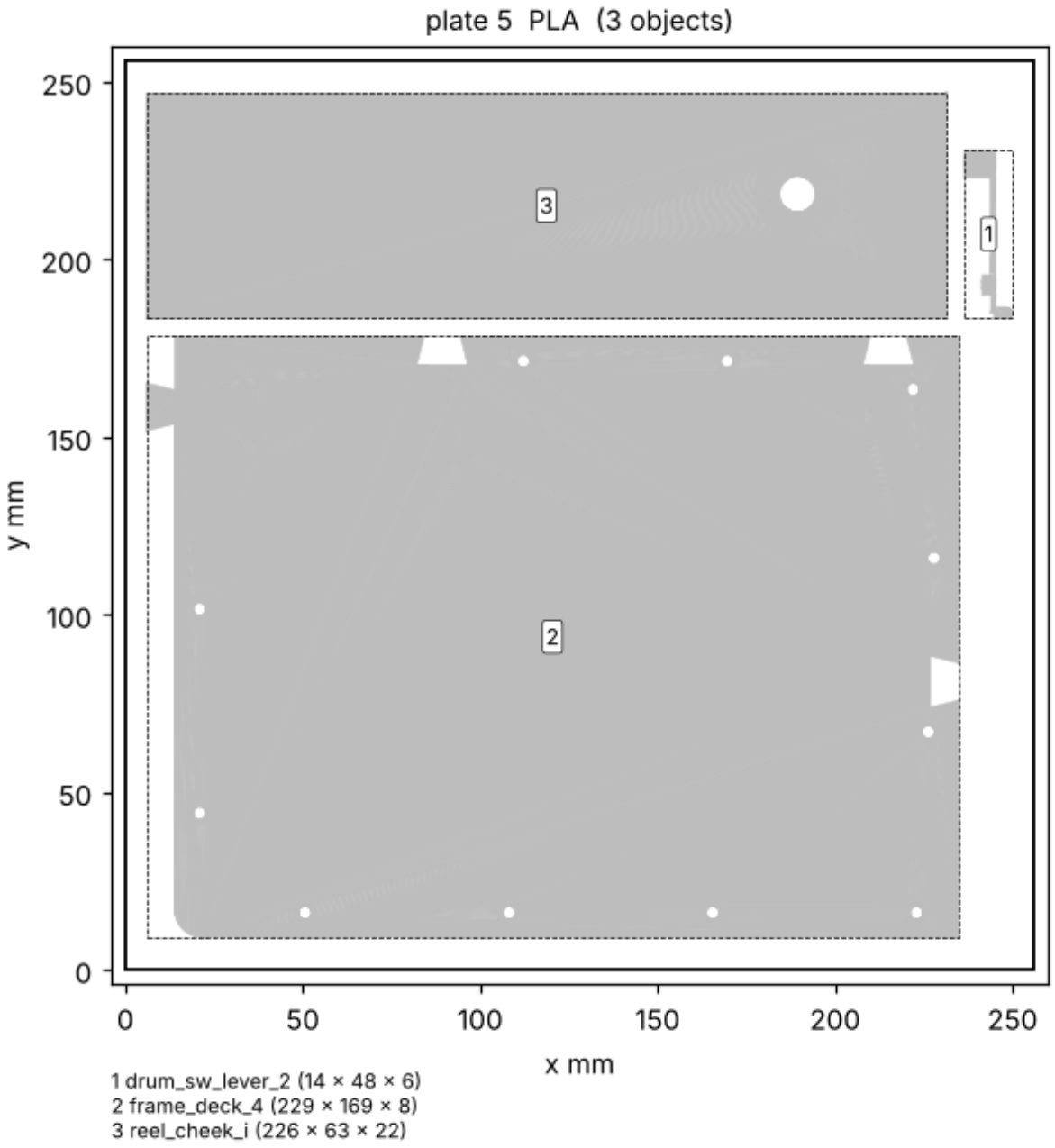


plate_5_PLA.3mf

PLA, 3 objects, about 353 g and 8.9 h.



1 drum_sw_lever_2 (14 × 48 × 6)
2 frame_deck_4 (229 × 169 × 8)
3 reel_cheek_i (226 × 63 × 22)

plate_6_PLA.3mf

PLA, 2 objects, about 343 g and 8.7 h.

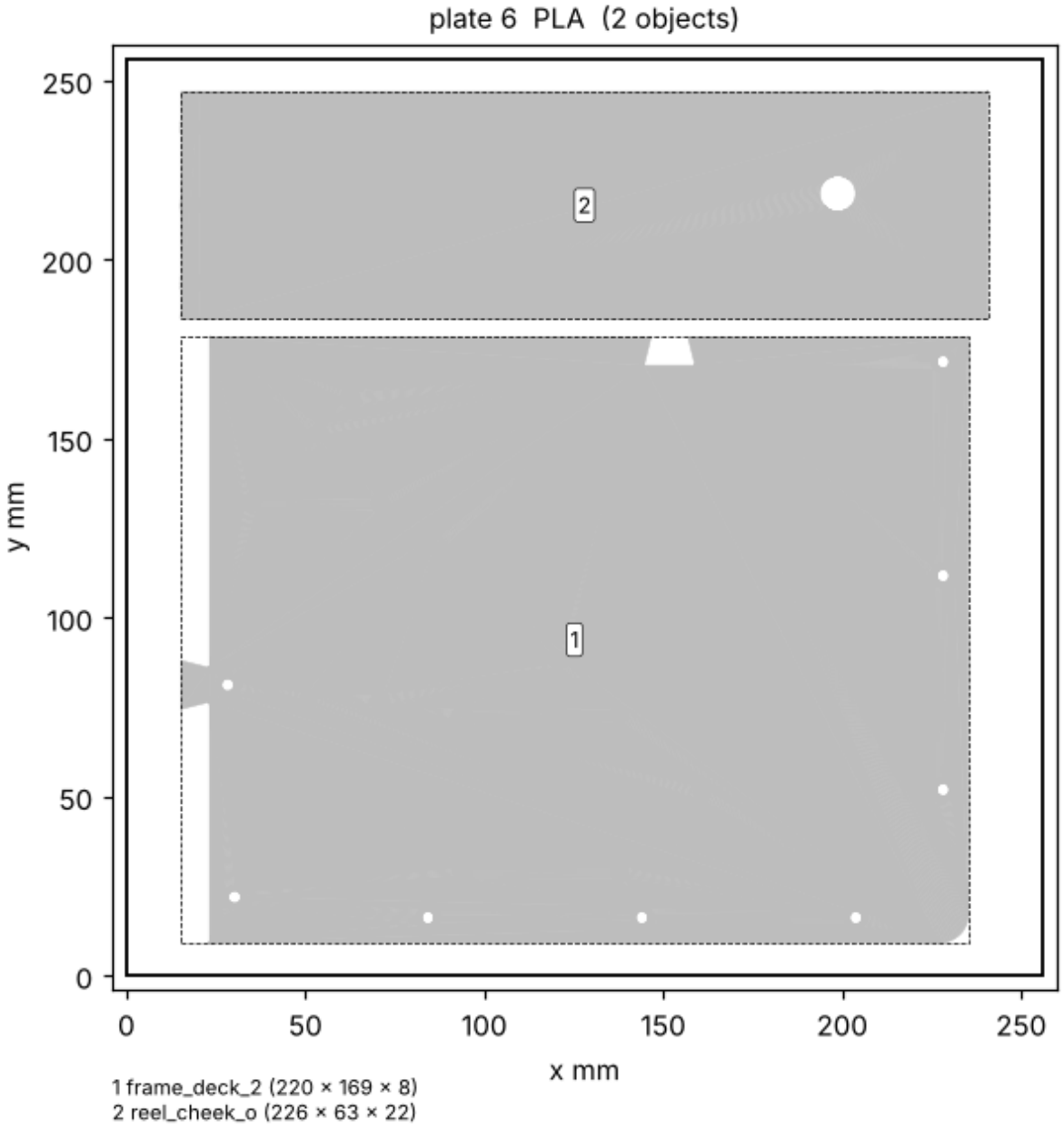


plate_7_PLA.3mf

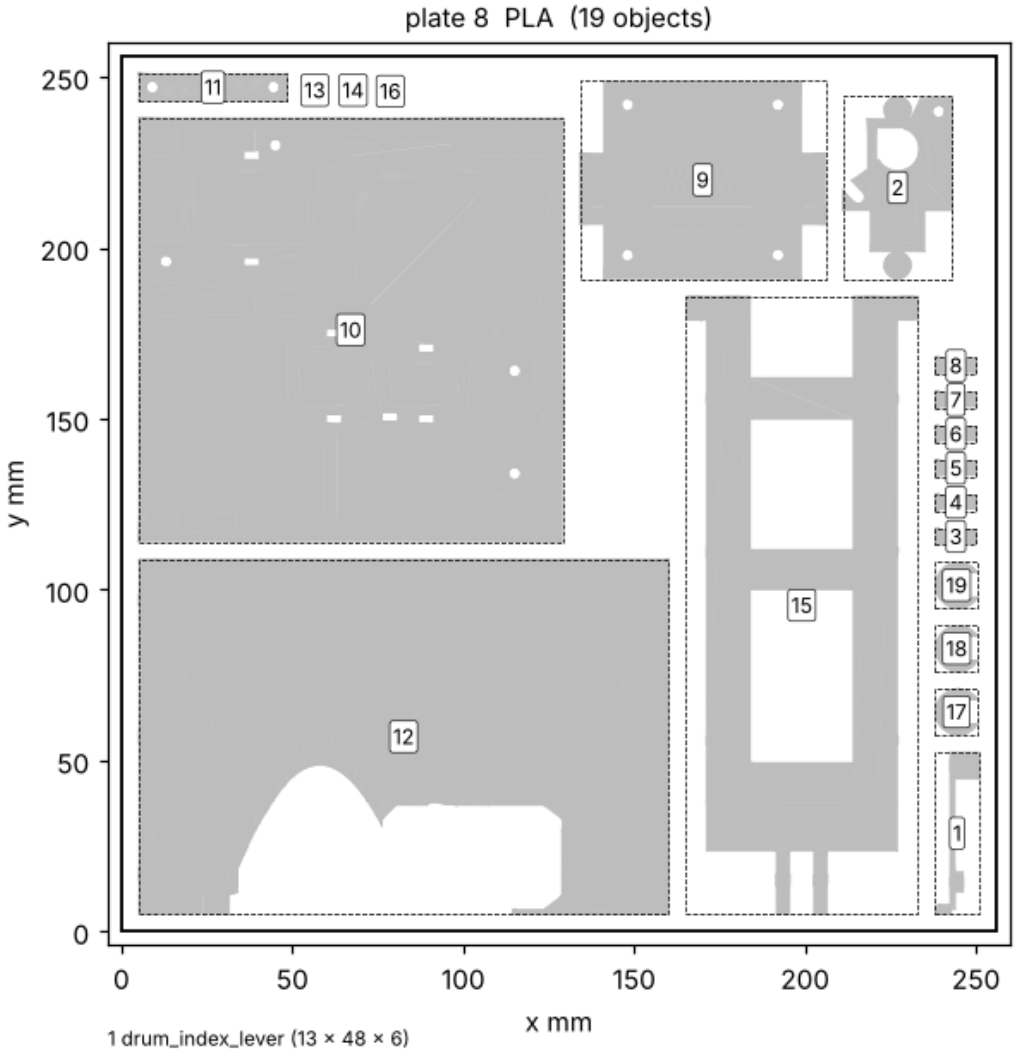
PLA, 13 objects, about 626 g and 17.1 h.



1 drum_dog_post (71 × 37 × 42)
2 drum_drum_a (97 × 97 × 75)
3 frame_rc832_holder (22 × 97 × 40)
4 launch_aim_crankpin_clip (10 × 11 × 1)
5 launch_aim_idler_clip (10 × 11 × 1)
6 launch_aim_lugpin_clip (10 × 11 × 1)
7 launch_rail_front (132 × 128 × 48)
8 launch_rail_splice_l (22 × 5 × 3)
9 launch_rail_splice_r (22 × 5 × 3)
10 reel_cs_clip (22 × 20 × 2)
11 reel_s_clip (10 × 11 × 1)
12 reel_sw_bracket (63 × 40 × 58)
13 reel_wall (82 × 201 × 213)

plate_8_PLA.3mf

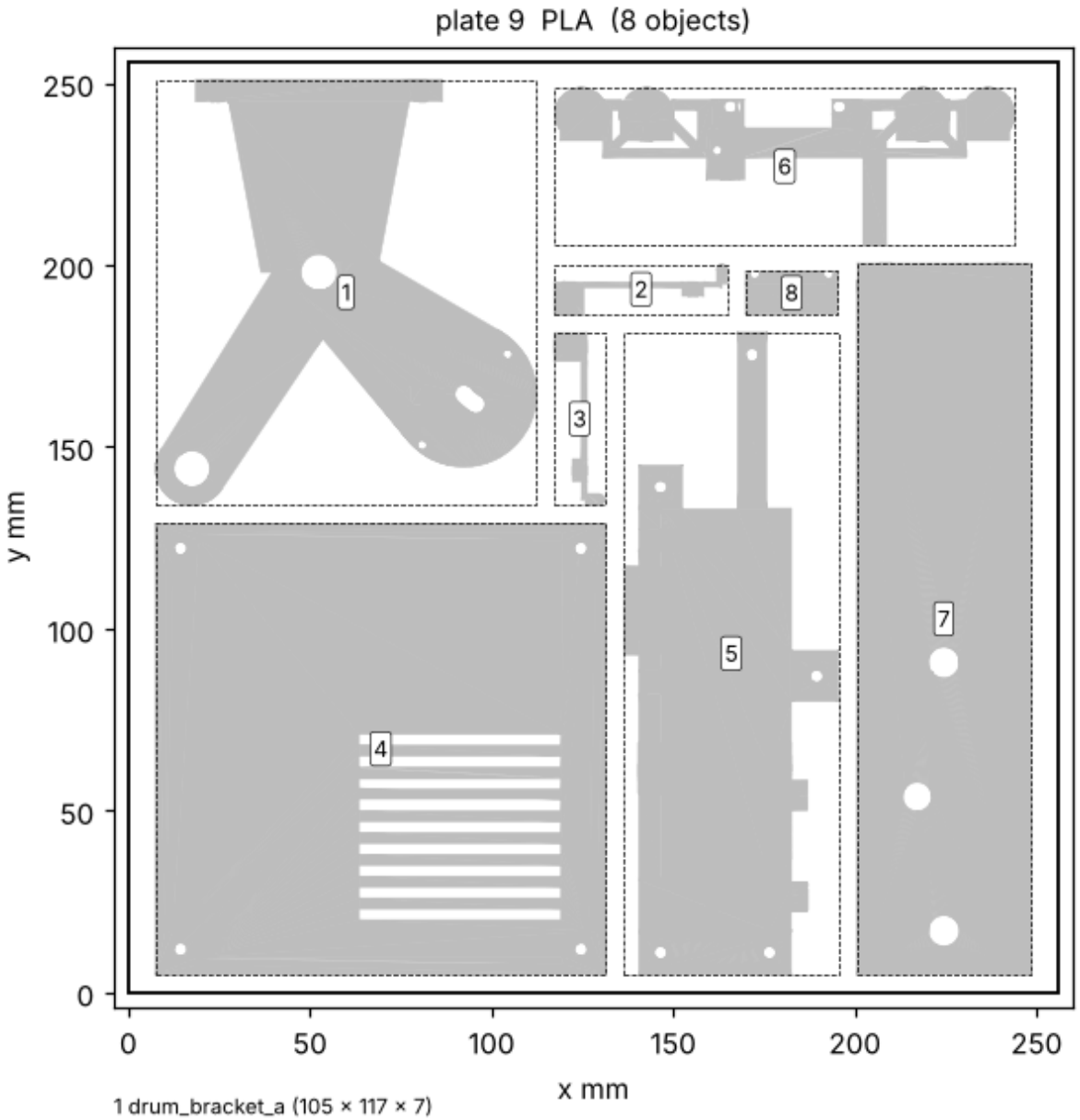
PLA, 19 objects, about 540 g and 15.1 h.



1 drum_index_lever (13 × 48 × 6)
2 drum_n20_cradle (32 × 54 × 19)
3 drum_shear_pin__c1 (12 × 5 × 4)
4 drum_shear_pin__c2 (12 × 5 × 4)
5 drum_shear_pin__c3 (12 × 5 × 4)
6 drum_shear_pin__c4 (12 × 5 × 4)
7 drum_shear_pin__c5 (12 × 5 × 4)
8 drum_shear_pin__c6 (12 × 5 × 4)
9 frame_camera_mount (72 × 58 × 236)
10 frame_ebay (124 × 124 × 58)
11 frame_ebay_esp_clamp (43 × 8 × 4)
12 frame_guard_front (155 × 104 × 16)
13 launch_muzzle_washer_l (6 × 6 × 1)
14 launch_muzzle_washer_r (6 × 6 × 1)
15 launch_rail_rear (68 × 181 × 76)
16 launch_seat_clip (6 × 6 × 2)
17 reel_k_clip (13 × 14 × 1)
18 reel_k_clip2 (13 × 14 × 1)
19 reel_kc_clip (13 × 14 × 1)

plate_9_PLA.3mf

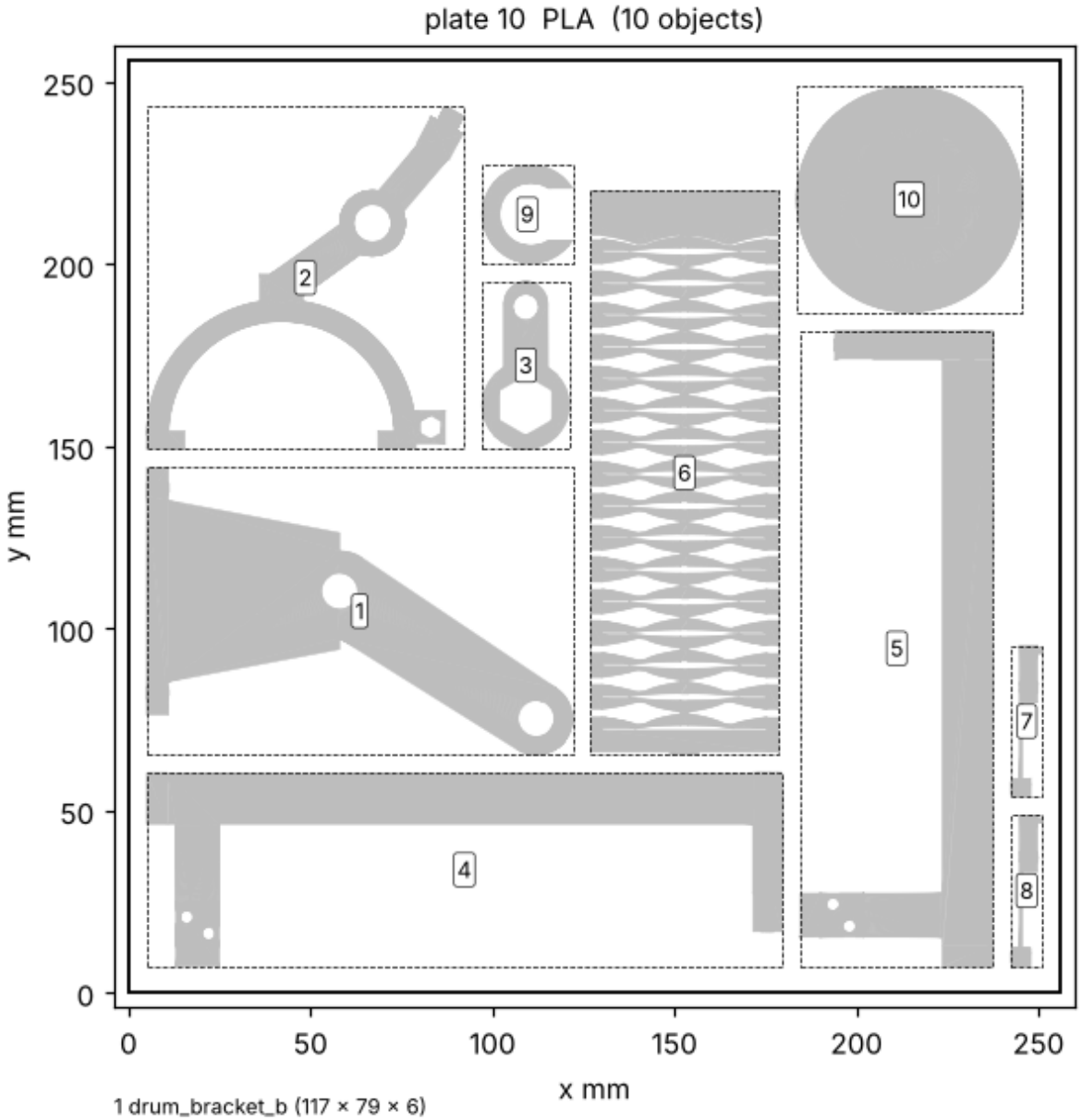
PLA, 8 objects, about 379 g and 10.7 h.



1 drum_bracket_a (105 × 117 × 7)
2 drum_sw_lever_0 (48 × 14 × 6)
3 drum_sw_lever_1 (14 × 48 × 6)
4 frame_ebay_lid (124 × 124 × 7)
5 launch_aim_housing (59 × 176 × 183)
6 launch_carriage_yoke (127 × 43 × 28)
7 reel_t_bracket (48 × 196 × 20)
8 sensors_as5600_strip_top (25 × 12 × 3)

plate_10_PLA.3mf

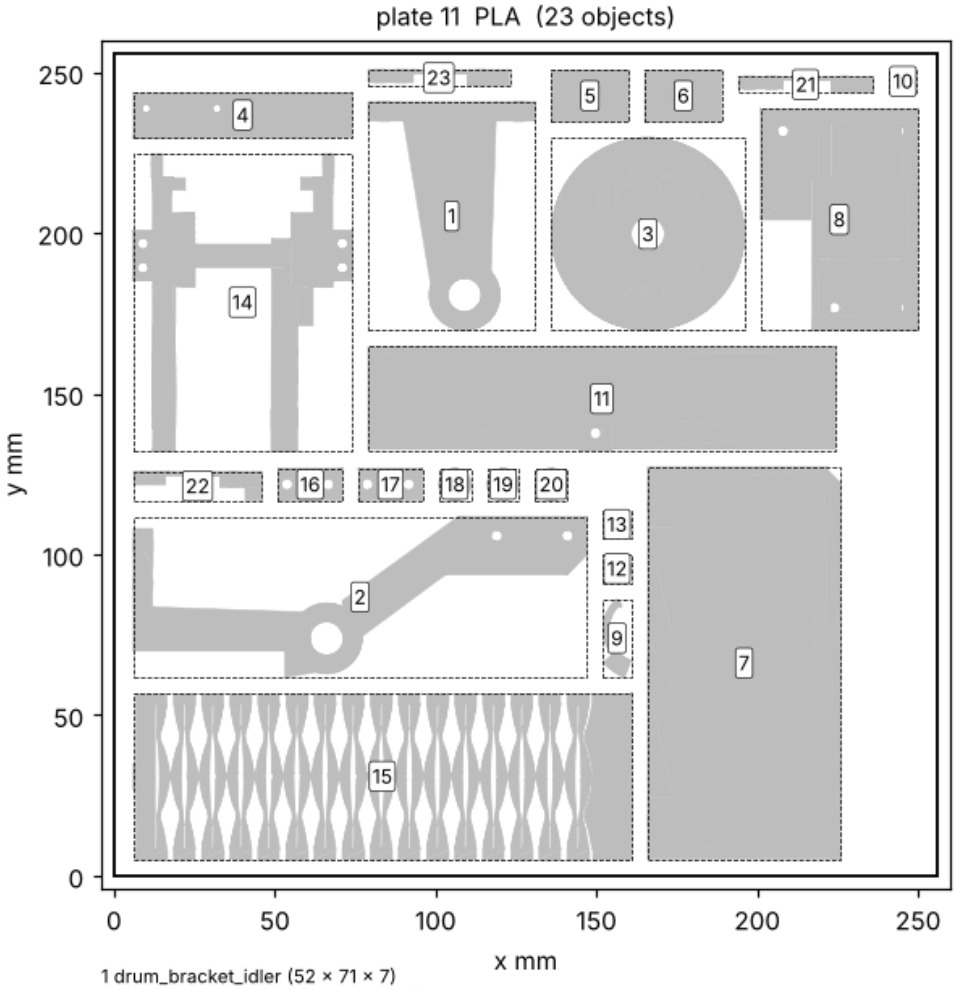
PLA, 10 objects, about 239 g and 6.5 h.



1 drum_bracket_b (117 × 79 × 6)
2 drum_fork (87 × 94 × 83)
3 drum_hand_lever (24 × 46 × 6)
4 launch_stop_frame_l (175 × 53 × 55)
5 launch_stop_frame_r (53 × 175 × 55)
6 launch_stop_spring_l (52 × 155 × 14)
7 launch_sw_home (8 × 41 × 6)
8 launch_sw_seated (8 × 42 × 6)
9 reel_sleeve_clip (25 × 27 × 3)
10 reel_spool (62 × 62 × 38)

plate_11_PLA.3mf

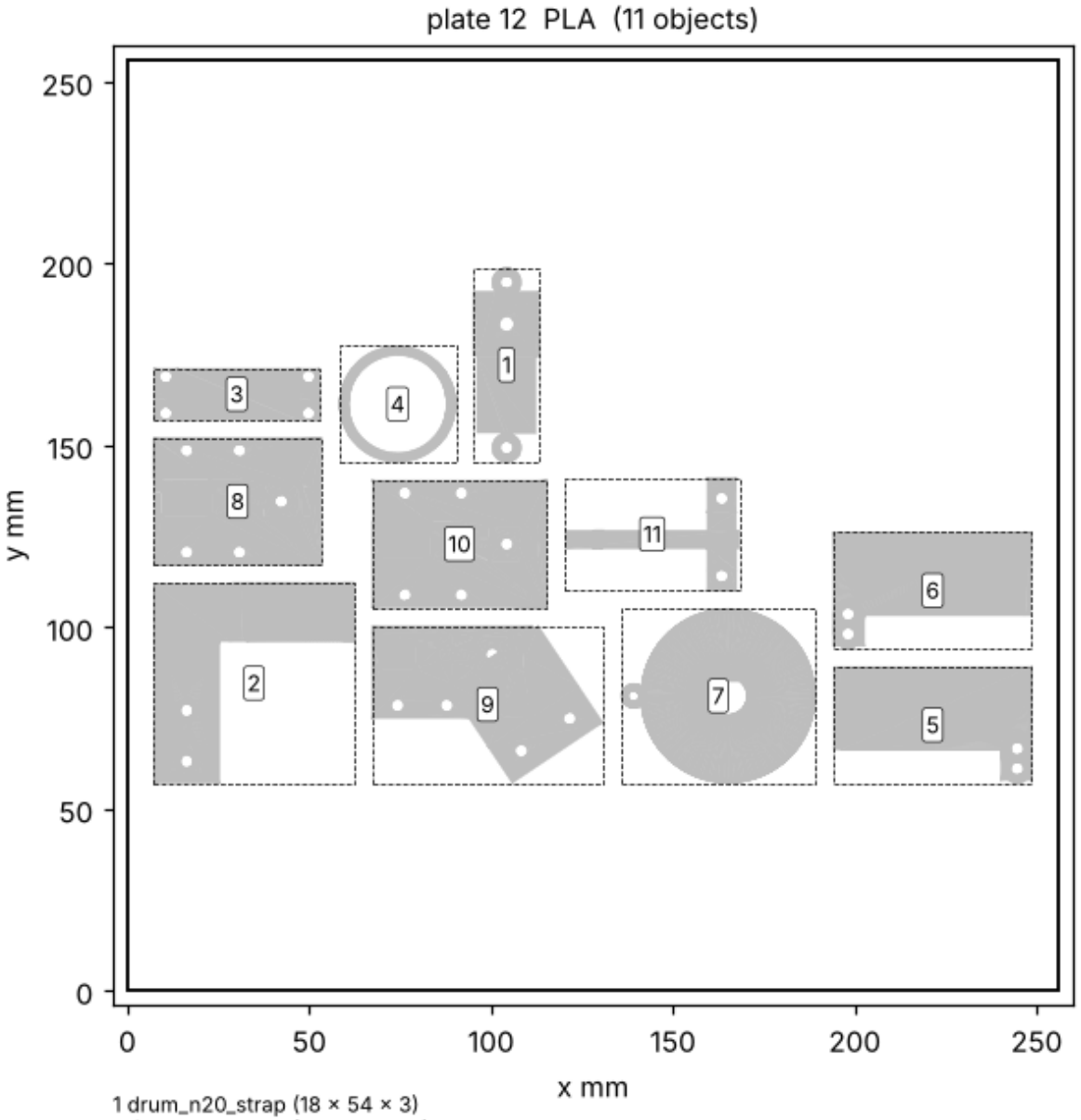
PLA, 23 objects, about 561 g and 15.3 h.



1 drum_bracket_idler (52 × 71 × 7)
2 drum_bracket_idler2 (141 × 50 × 6)
3 drum_drum_b (60 × 60 × 58)
4 drum_index_block (68 × 14 × 8)
5 frame_cable_anchor_1 (24 × 16 × 10)
6 frame_cable_anchor_2 (24 × 16 × 10)
7 frame_guard_hood (60 × 122 × 14)
8 frame_motor_leg (49 × 69 × 187)
9 launch_aim_drag_pad (9 × 24 × 4)
10 launch_aim_drag_shim (9 × 7 × 4)
11 launch_guard_l_b2 (145 × 32 × 40)
12 launch_pulley_top_washer_l (9 × 9 × 2)
13 launch_pulley_top_washer_r (9 × 9 × 2)
14 launch_rail_mid (68 × 92 × 170)
15 launch_stop_spring_r (155 × 52 × 14)
16 reel_bar_cap_1 (20 × 10 × 5)
17 reel_bar_cap_2 (20 × 10 × 5)
18 reel_spacer_1 (10 × 10 × 34)
19 reel_spacer_2 (10 × 10 × 34)
20 reel_spacer_3 (10 × 10 × 34)
21 reel_sw_lever (42 × 5 × 8)
22 sensors_sw_gap_lever (40 × 9 × 8)
23 sensors_sw_rail_low_lever (44 × 5 × 8)

plate_12_PLA.3mf

PLA, 11 objects, about 101 g and 3.0 h.



1 drum_n20_strap (18 × 54 × 3)
2 frame_can_stay (55 × 55 × 71)
3 launch_aim_cap (46 × 14 × 19)
4 launch_drive_washer (32 × 32 × 1)
5 launch_guard_l_c (54 × 32 × 26)
6 launch_guard_r_c (54 × 32 × 26)
7 reel_cup (53 × 48 × 18)
8 reel_sw_base (46 × 35 × 12)
9 sensors_sw_gap (63 × 43 × 12)
10 sensors_sw_rail_low (48 × 35 × 12)
11 sensors_sw_rail_low_bracket (48 × 31 × 74)

plate_13_PLA.3mf

PLA, 1 objects, about 340 g and 8.5 h.

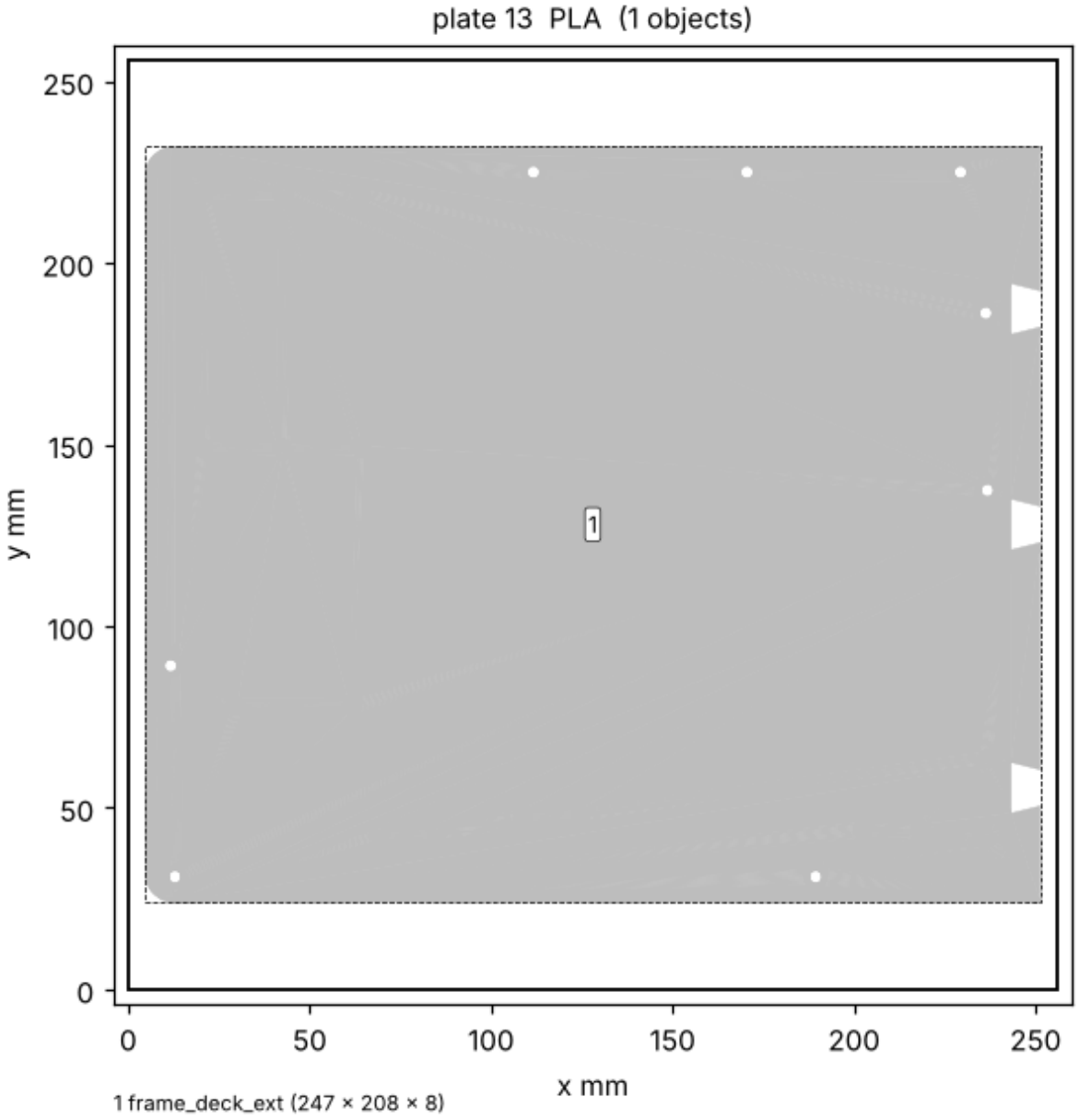


plate_14_PLA.3mf

PLA, 1 objects, about 65 g and 1.9 h.

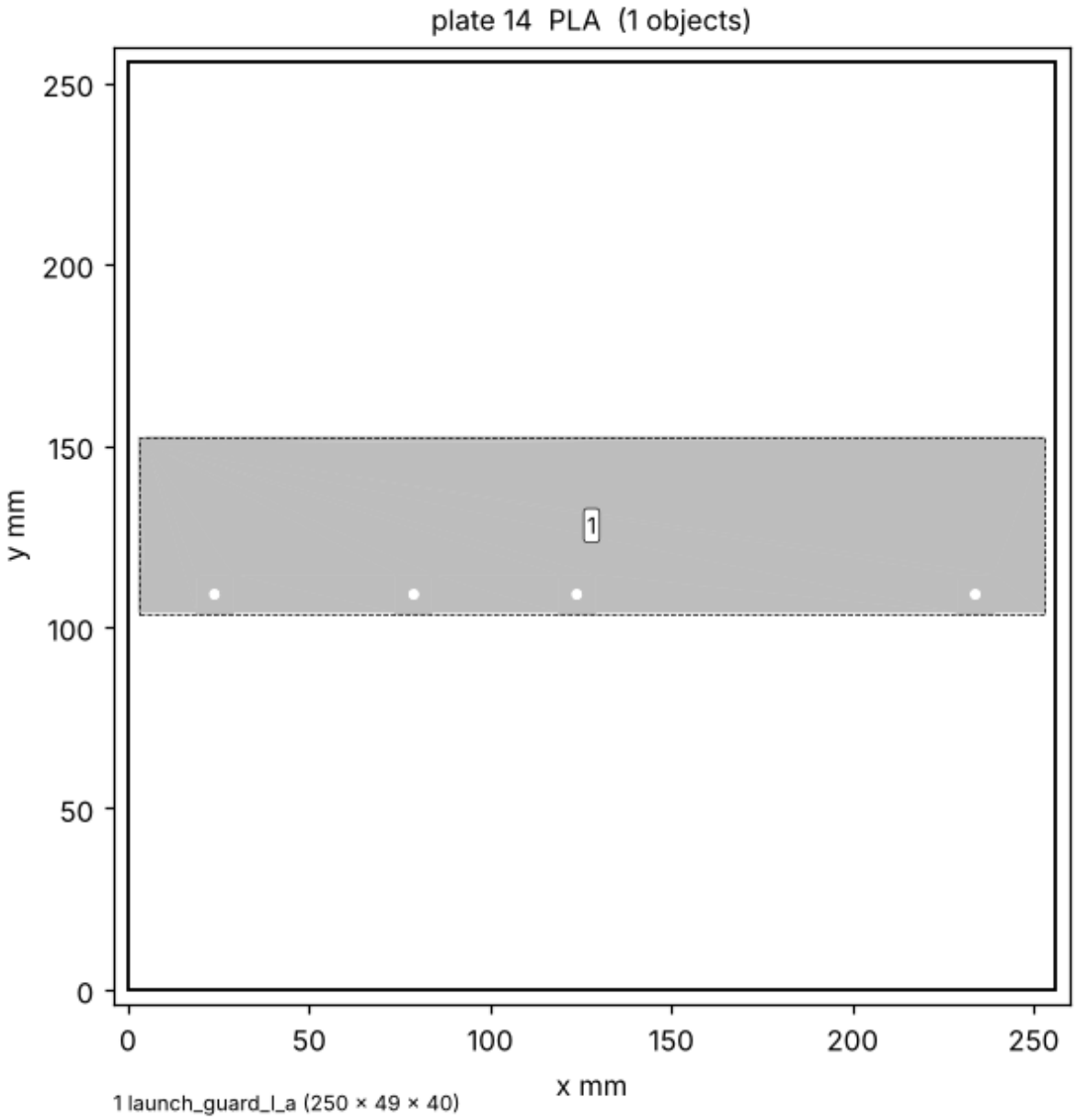


plate_15_PLA.3mf

PLA, 1 objects, about 65 g and 1.9 h.

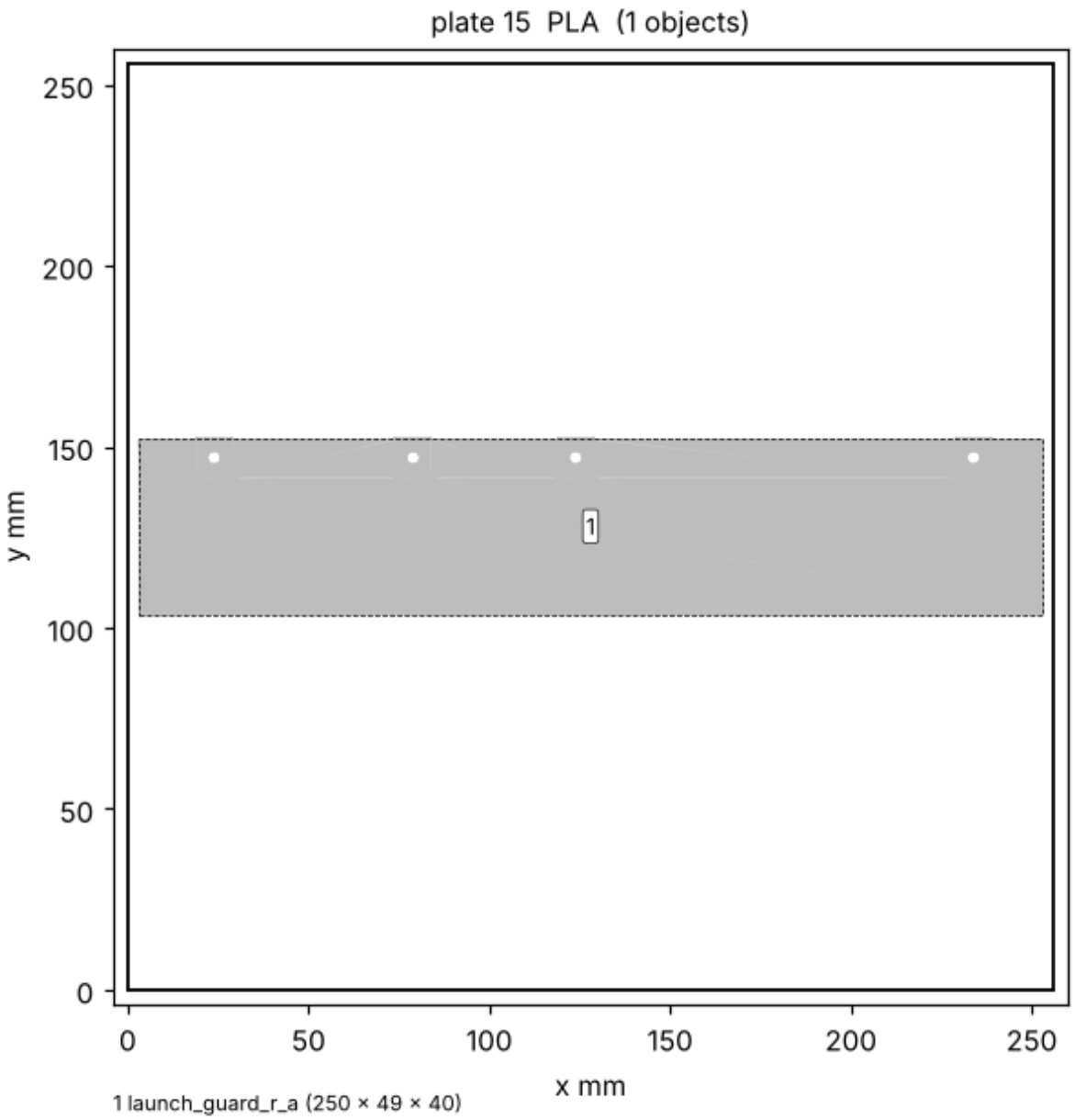
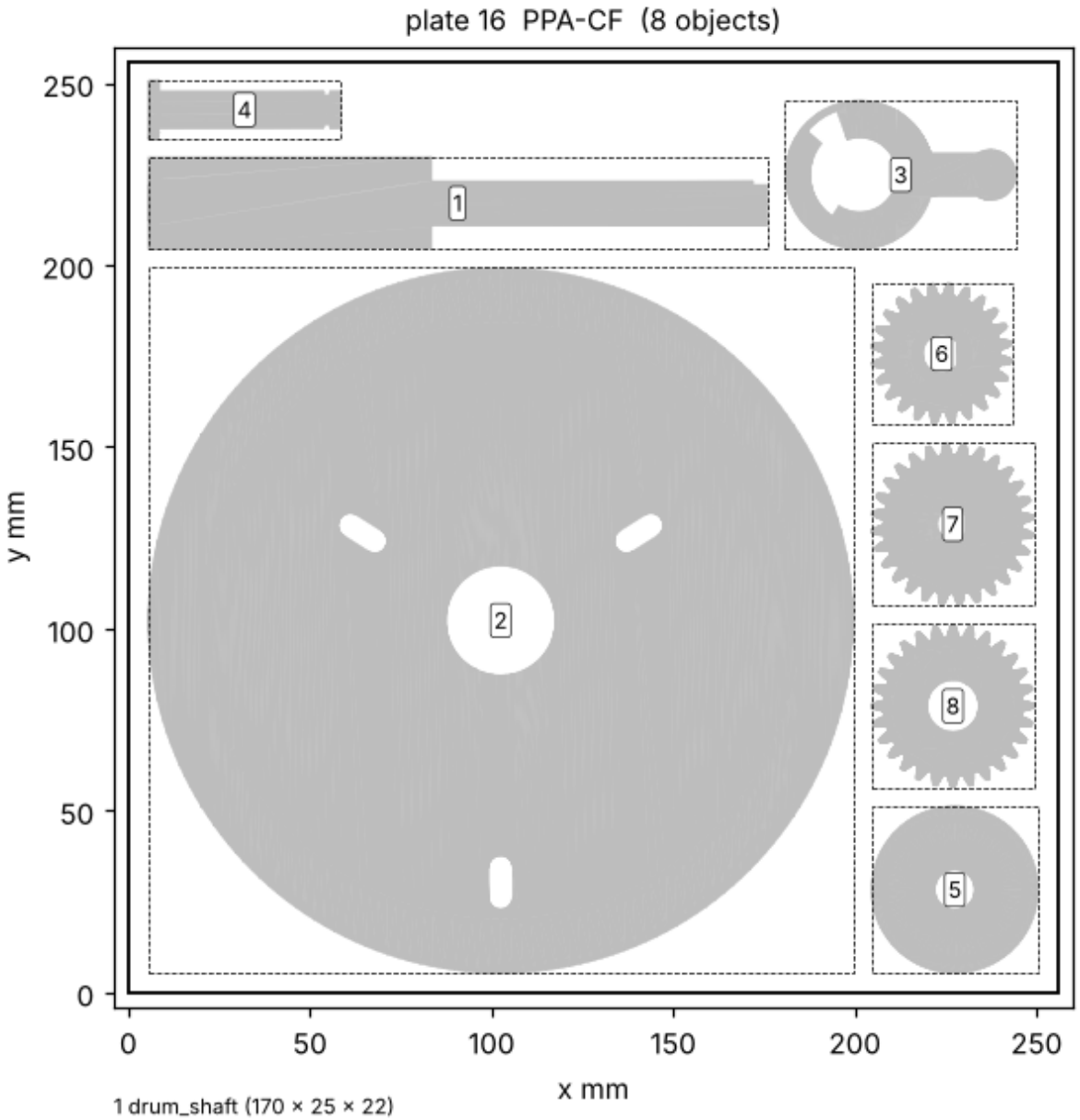


plate_16_PPA-CF.3mf

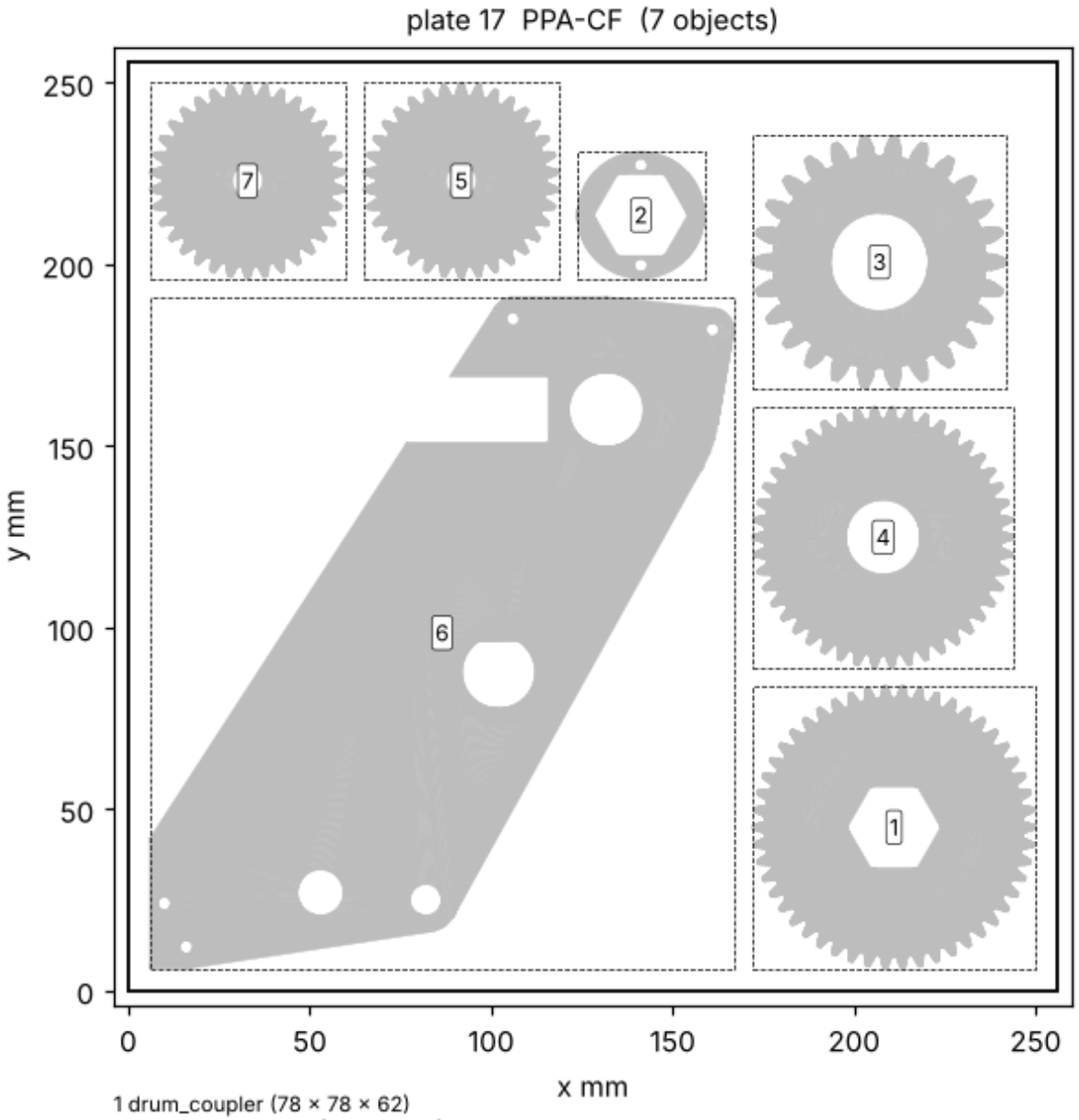
PPA-CF, 8 objects, about 436 g and 19.7 h.



1 drum_shaft (170 × 25 × 22)
2 frame_motor_adapter (194 × 194 × 16)
3 launch_aim_arm (64 × 41 × 19)
4 launch_gear_pin (53 × 16 × 12)
5 reel_drum (46 × 46 × 43)
6 reel_gk (39 × 39 × 23)
7 reel_gt2 (45 × 45 × 6)
8 reel_hub (45 × 45 × 17)

plate_17_PPA-CF.3mf

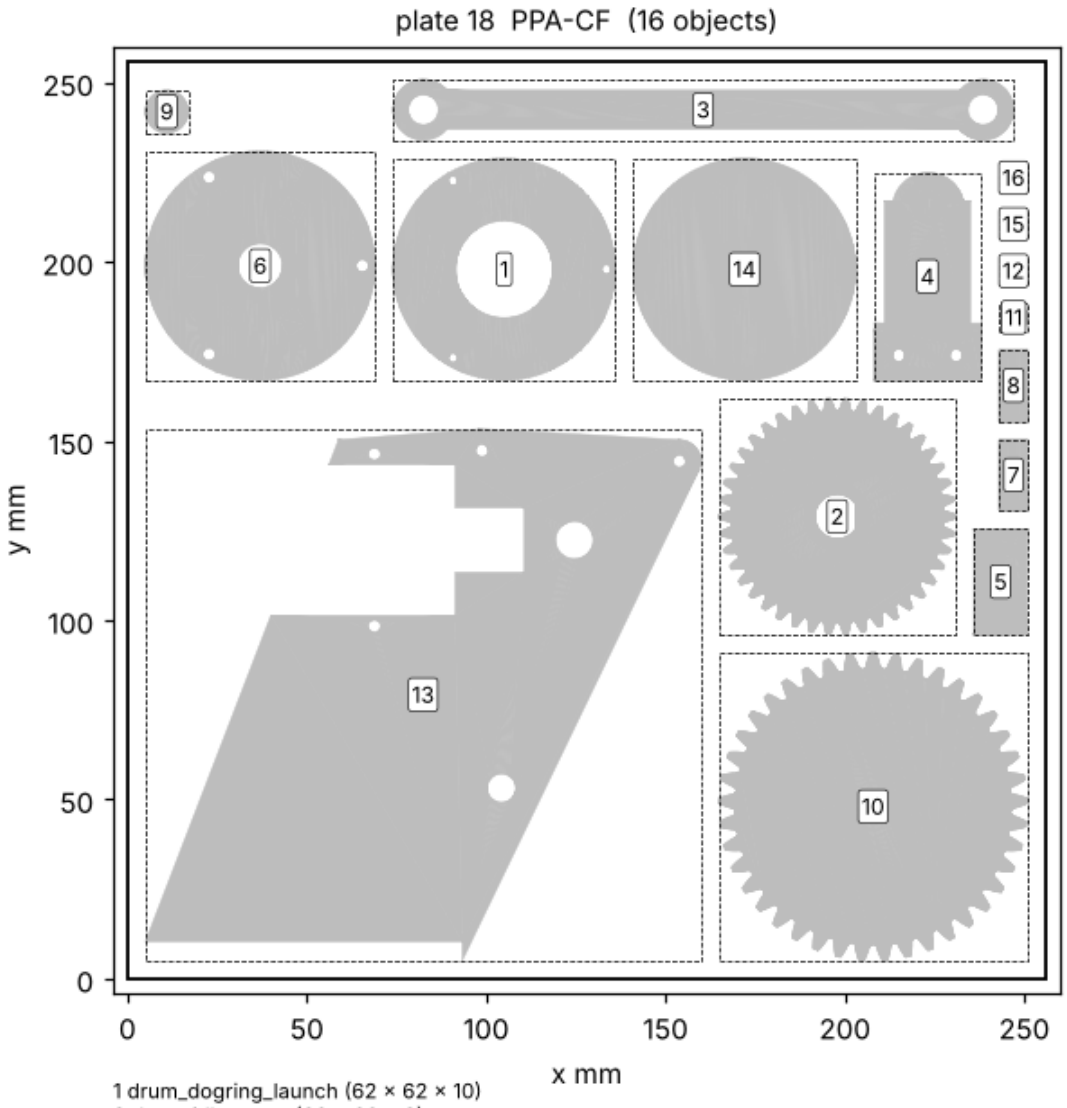
PPA-CF, 7 objects, about 285 g and 13.1 h.



1 drum_coupler (78 × 78 × 62)
2 drum_reel_washer (35 × 35 × 1)
3 launch_drive_gear (70 × 70 × 29)
4 reel_bus (72 × 72 × 10)
5 reel_ik1 (54 × 54 × 5)
6 reel_p1 (161 × 185 × 14)
7 reel_ti (54 × 54 × 6)

plate_18_PPA-CF.3mf

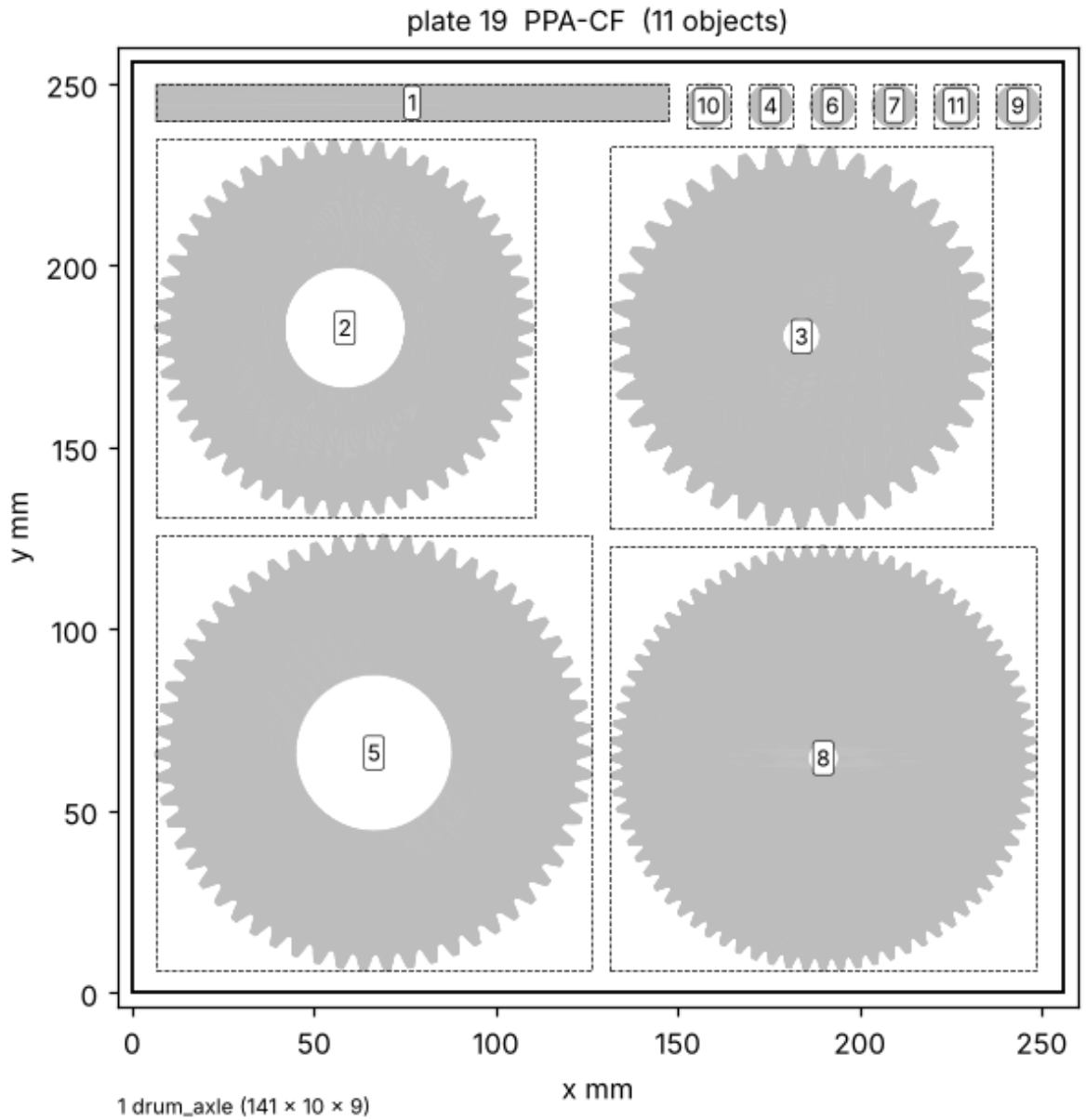
PPA-CF, 16 objects, about 337 g and 16.6 h.



1 drum_dogring_launch (62 × 62 × 10)
2 drum_idler_gear (66 × 66 × 9)
3 launch_aim_conrod (173 × 17 × 6)
4 launch_aim_lug (30 × 58 × 28)
5 launch_seat_plunger (15 × 30 × 20)
6 launch_sleeve (64 × 64 × 85)
7 launch_yoke_peg_l (8 × 20 × 7)
8 launch_yoke_peg_r (8 × 20 × 7)
9 reel_crank_washer (12 × 12 × 1)
10 reel_ic (86 × 86 × 12)
11 reel_lane_k (8 × 8 × 234)
12 reel_lane_t (8 × 8 × 203)
13 reel_p2 (155 × 148 × 6)
14 reel_s (62 × 62 × 48)
15 reel_ss (8 × 8 × 157)
16 reel_t2 (8 × 8 × 39)

plate_19_PPA-CF.3mf

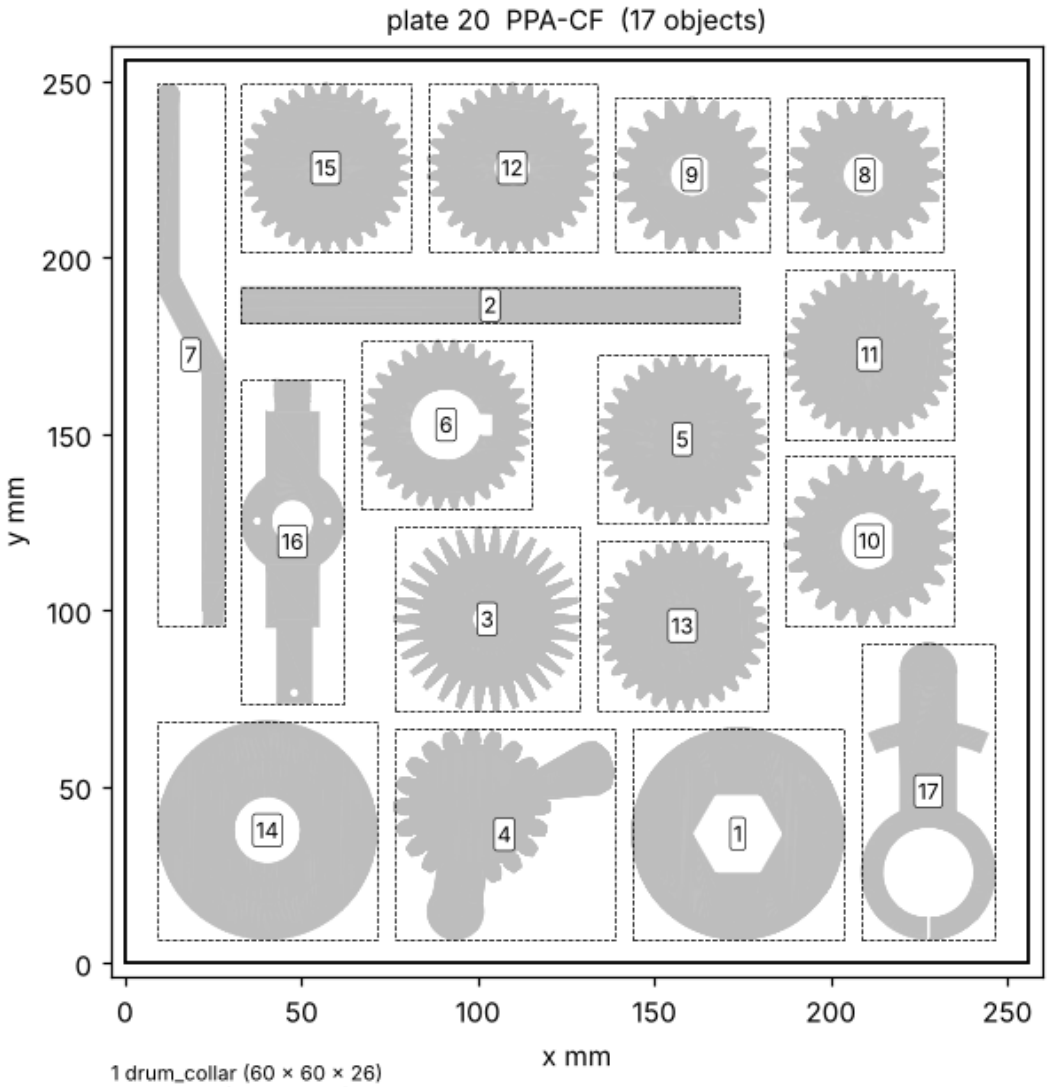
PPA-CF, 11 objects, about 466 g and 21.9 h.



1 drum_axle (141 × 10 × 9)
2 drum_gear_reel (104 × 104 × 19)
3 launch_big_gear (105 × 105 × 27)
4 reel_ic_axle (12 × 12 × 31)
5 reel_icr (120 × 120 × 14)
6 reel_ik1_axle (12 × 12 × 18)
7 reel_ik2_axle (12 × 12 × 18)
8 reel_ikr (117 × 117 × 5)
9 reel_ikr_axle (12 × 12 × 8)
10 reel_lane_c (12 × 12 × 161)
11 reel_ti_axle (12 × 12 × 15)

plate_20_PPA-CF.3mf

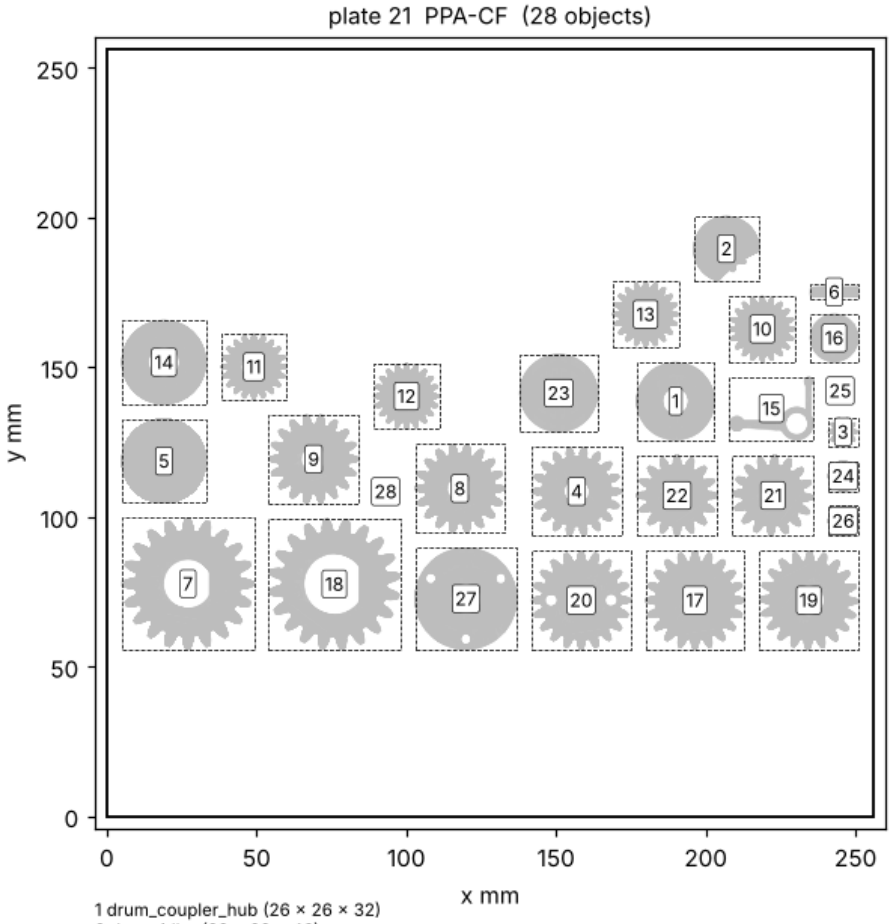
PPA-CF, 17 objects, about 249 g and 11.5 h.



1 drum_collar (60 × 60 × 26)
2 drum_rail (141 × 10 × 9)
3 launch_aim_bevel30 (53 × 52 × 10)
4 launch_aim_crank (62 × 60 × 38)
5 launch_aim_output (48 × 48 × 60)
6 launch_aim_sun (48 × 48 × 12)
7 reel_bar (19 × 154 × 16)
8 reel_bcx (44 × 44 × 8)
9 reel_ga (44 × 44 × 8)
10 reel_gc (48 × 48 × 12)
11 reel_gi (48 × 48 × 6)
12 reel_gk1 (48 × 48 × 5)
13 reel_gk2 (48 × 48 × 5)
14 reel_hub_flange (62 × 62 × 2)
15 reel_ik2 (48 × 48 × 5)
16 reel_rocker (29 × 92 × 6)
17 reel_swing_arm (38 × 84 × 9)

plate_21_PPA-CF.3mf

PPA-CF, 28 objects, about 168 g and 8.2 h.



1 drum_coupler_hub (26 × 26 × 32)
2 drum_idler (22 × 22 × 46)
3 drum_n20_pinion (10 × 9 × 8)
4 launch_aim_idler (30 × 30 × 12)
5 launch_aim_worm (28 × 28 × 54)
6 launch_seat_pushrod (16 × 5 × 4)
7 reel_bc (44 × 44 × 8)
8 reel_bk2 (30 × 30 × 8)
9 reel_bkx (30 × 30 × 7)
10 reel_bt2 (22 × 22 × 7)
11 reel_bt3 (22 × 22 × 6)
12 reel_bt3x (22 × 22 × 6)
13 reel_btx (22 × 22 × 6)
14 reel_c (28 × 28 × 65)
15 reel_crank (28 × 21 × 12)
16 reel_cs (16 × 16 × 43)
17 reel_ga2 (33 × 33 × 5)
18 reel_gc2 (44 × 44 × 8)
19 reel_gkc (33 × 33 × 5)
20 reel_gr (33 × 33 × 6)
21 reel_gs (27 × 27 × 6)
22 reel_gt3 (27 × 27 × 6)
23 reel_icr_axle (26 × 26 × 33)
24 reel_k (10 × 10 × 59)
25 reel_k2 (8 × 8 × 33)
26 reel_kc (10 × 10 × 103)
27 reel_sleeve (34 × 34 × 24)
28 reel_t3 (8 × 8 × 21)